A JAX+PETSc Toolset for Nonlinear Finite-Element Energy Minimization:
Local Automatic Differentiation, Sparse Assembly, and Solver Policy

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Abstract

Large nonlinear finite-element energy problems require reliable local derivatives, robust nonlinear globalization, and sparse distributed linear algebra that is matched to the mechanics of the problem. We develop a scientific toolset for studying these requirements as coupled numerical choices in nonlinear finite-element energy minimization and energy-based topology optimization. The primary JAX+PETSc realization uses JAX for local energy and constitutive evaluations and PETSc for sparse assembly, Newton globalization, Krylov solvers, and preconditioners. The benchmark suite covers p-Laplace, Ginzburg–Landau, finite-strain hyperelasticity, two- and three-dimensional Mohr–Coulomb plasticity, and topology optimization.

On representative fixed-work cases, we compare element automatic differentiation, constitutive automatic differentiation, and colored sparse recovery for constructing second-order information, and measure their interaction with PETSc assembly, nonlinear globalization, Krylov solvers, and preconditioners. In the fixed high-order three-dimensional plasticity derivative-route case, constitutive AD has the lowest measured wall time among the tested routes while preserving endpoint energy, external-work, and maximum-displacement observables. The distributed mechanics examples give MPI timing and solver-policy evidence, and the topology example tests compliance and density consistency across MPI ranks under a controlled fixed-iteration protocol. Scoped comparators supply context: matched pure JAX and FEniCS formulations provide internal checks where available, JAX-FEM gives a narrow hyperelastic endpoint-state comparison, the Sysala-family works provide plasticity reference-model context, and the Čermák–Sysala–Valdman MATLAB work provides implementation context rather than a numerical baseline. Together, the evidence indicates that, for the tested problem classes, derivative construction should be evaluated inside the sparse nonlinear solve that consumes it, together with assembly, globalization, Krylov algebra, and preconditioning.

1 Introduction

Nonlinear finite-element models in mechanics, scalar PDEs, and design optimization often combine three computational difficulties. The local model may be easiest to write as an energy or constitutive law whose derivatives are tedious to assemble by hand. The global problem then requires robust nonlinear globalization and sparse Newton–Krylov linear algebra. Finally, the implementation must remain comparable across reference formulations, derivative approximations, and distributed-memory solver choices. A useful scientific toolset for this class of problems should therefore expose not only a modeling interface, but also the derivative route, assembly layout, nonlinear solver policy, and preconditioner design.

We develop and evaluate such a scientific toolset for nonlinear finite-element energy minimization and optimization. In the primary JAX+PETSc realization, JAX defines local energies and constitutive laws, while PETSc provides sparse distributed algebra, nonlinear globalization, Krylov methods, and preconditioners. We use comparison evidence in scoped roles: internal formulation checks, a narrow endpoint companion comparison, plasticity reference-model context, and legacy MATLAB implementation context. The pure JAX and FEniCS formulations provide internal checks where matched benchmark definitions exist. JAX-FEM [1] supplies the hyperelastic endpoint-state comparison. The Sysala-family slope-stability literature [2–5] provides reference-model context for the plasticity endpoint observables. The Čermák–Sysala–Valdman MATLAB work [6] is used as elastoplastic implementation context rather than as a numerical baseline.

1.1 Contributions

Our contribution is a JAX+PETSc realization and numerical study that treats derivative construction, MPI sparse assembly, nonlinear globalization, and preconditioning as coupled design choices for nonlinear FEM energy minimization. The central thesis is that derivative construction should be evaluated inside the sparse nonlinear solve that consumes it. The study also encodes solver expertise through controlled selection and reporting of derivative routes, assembly ownership, globalization, Krylov tolerances, and preconditioners under fixed benchmark definitions. We support this thesis through three technical components:

1. **An integrated derivative–assembly path.** We show how local JAX energy or constitutive evaluations can feed PETSc-owned sparse MPI residual, tangent, and Hessian-vector-product computations without changing the benchmark functional being solved.

2. **Controlled derivative-route comparisons.** We compare element automatic differentiation (AD), constitutive AD, and colored sparse recovery inside the same solver realization, so the discrete model is held fixed while derivative-route and associated assembly or recovery costs are compared.

3. **Solver-policy evidence across model classes.** We study nonlinear globalization choices, Krylov and preconditioner choices, and multigrid or coarse-solver choices across selected scalar, mechanics, plasticity, and topology cases, with conclusions restricted to the tested benchmark definitions.

Relative to the JAX-native differentiable FEM, FEM-adjoint, and bridge architectures discussed in Section 2, our emphasis is the integrated evaluation of local AD routes inside PETSc-owned sparse MPI Newton solvers, with globalization and preconditioner policy varied on the heterogeneous benchmark suite studied here.

We evaluate these components on six benchmark families: p-Laplace, scalar Ginzburg–Landau, hyperelasticity, two- and three-dimensional Mohr–Coulomb-type plasticity, and topology optimization. The scalar problems isolate formulation and derivative agreement. Hyperelasticity adds finite-strain mechanics with a matched JAX-FEM endpoint-state companion comparison. The plasticity benchmarks test nonsmooth constitutive response, endpoint-observable comparison, and multigrid behavior. Topology optimization tests a distributed design-and-mechanics loop. The geomechanics context follows the finite-element strength-reduction and 3D slope-stability literature [2–5, 7], but the present plasticity study is limited to final-load endpoint observables rather than path-consistent incremental plastic-history validation.

The surrounding literature covers expressive finite-element automation, differentiable FEM, JAX–solver bridges, mechanics and topology benchmark families, and sparse nonlinear solver infrastructure. The present work combines these roles by treating local differentiation, PETSc-owned sparse MPI assembly, nonlinear globalization, and preconditioner policy in one heterogeneous benchmark suite. Section 2 discusses these source classes by technical role.

The paper is organized as follows. Sections 2 to 4 position the related work, define the common finite-element and derivative notation, and describe the primary JAX+PETSc realization. Section 5 defines the benchmark suite, Section 6 separates validation and external-comparison evidence, and Section 7 reports timing, scaling, derivative-route, and discretization evidence. Sections 8 and 9 synthesize the scope and limitations; supporting solver-policy diagnostics are collected in Section A.

2 Related Work

High-level finite-element systems automate variational forms and adjoint workflows, JAX-native finite-element codes automate local differentiation, and PETSc supplies scalable sparse nonlinear algebra. For nonlinear energy minimization, these capabilities meet inside the assembled Newton solve: the derivative route, ownership layout, globalization, Krylov method, and preconditioner are coupled numerical choices. The relevant literature therefore separates into high-level FEM automation, differentiable FEM and JAX–solver bridges, mechanics and topology benchmark context, and sparse nonlinear algebra. The focus of the present study is their combined evaluation on one heterogeneous energy-minimization suite.

The sources summarized below treat these ingredients in different combinations. The comparison design used here combines matched derivative-route comparisons inside PETSc-owned sparse MPI Newton solves with separated validation, endpoint-comparison, timing, and solver-policy claims. The table therefore records roles, not rankings.

Table 1 summarizes the computational roles needed for this positioning: modeling layer, differentiation scope, second-order information, parallel execution model, and the closest benchmark families.

Finite-element and PDE-optimization automation. The original FEniCS project established the high-level finite-element programming model in which weak forms are expressed symbolically and compiled into efficient kernels [8]. The more recent `DOLFINx` project keeps that mathematical abstraction while moving to a data-oriented design with explicit support for modern parallelism, custom kernels, and direct access to lower-level data structures [9]. In PDE-constrained optimization, `dolfin-adjoint` showed that adjoints of high-level transient finite-element programs can be derived automatically from the symbolic problem description [10], and `pyadjoint` documents the FEniCS/Firedrake adjoint framework used to automate such computations in practice [11]. The `cashocs` software paper extends this PDE-optimization thread to shape optimization, optimal control, topology optimization via level sets, and MPI-enabled computations [12]. These works are the natural reference point for expressive finite-element and adjoint-centered automation. The present paper uses that context while focusing instead on second-order routes, nonlinear globalization, and sparse distributed solver policy. In the present evidence surface, FEniCS-style formulations are used as matched reference formulations where available, not as broad performance baselines.

Table 1: Selected related computational roles for the software and literature families most relevant to this paper.

Technical family	Documented role	Relation to this work
FEM and adjoint automation [8–12]	High-level variational forms, generated kernels, adjoint-based sensitivities, and PDE-optimization loops.	Provides the reference high-level FEM and optimization context; selected FEniCS formulations are scoped comparisons.
JAX-native differentiable FEM and PDE solvers [1, 13–15]	Source-specific roles: GPU-oriented differentiable mechanics (JAX-FEM/JAX-CPFEM), implicit differentiation and solver options (AutoPDEx), and Hessian-vector inverse-problem studies (Xue 2026).	Motivates local differentiable modeling; the present study instead couples local JAX derivatives to PETSc sparse MPI solves and compares derivative routes.
FEM–JAX and JAX–PETSc bridge architectures [16–18]	Source-specific bridges: Firedrake–JAX variational coupling, FEniCSx external operators for constitutive models, and JetSCI-style JAX–PETSc differentiated local discretizations.	Architectural context for the JAX+PETSc realization; this study evaluates derivative construction, globalization, and preconditioning on the stated benchmark suite.
Slope-stability and elastoplastic implementation context [2–7]	Strength-reduction plasticity, nonsmooth return mapping, continuation, and practical 2D/3D elastoplastic implementation.	Sysala-family works provide slope-stability reference-model context; Čermák–Sysala–Valdman provides practical elastoplastic implementation context. Numerical claims remain limited to the implemented endpoint surrogate and reported comparison observables.
Topology-optimization benchmark lineage [19–23]	SIMP compliance minimization, density filtering, compact educational implementations, and modern parallel topology software.	Defines the topology problem context; the numerical claims are the reported design-and-mechanics timing and rank-consistency diagnostics.

Differentiable FEM and JAX–solver bridges. The broad automatic-differentiation (AD) background is surveyed by Baydin et al. [24], while JAX provides a program-transformation environment for higher-order derivatives, vectorization, and just-in-time compilation in scientific computing [25]. Within finite elements, JAX-FEM demonstrates inverse design, nonlinear constitutive modeling, and GPU-oriented differentiable computations in a JAX-native 3D FE solver [1]. Xue's later finite-element differentiable-physics paper is a close methodological reference for second-order information because it derives implicit Hessian-vector products and evaluates Newton–CG use of exact Hessian information on finite-element inverse-problem benchmarks [13]. AutoPDEx broadens the JAX-native PDE picture with nonlinear minimizers, implicit differentiation, and optional PETSc integration [14]; JAX-CPFEM adds differentiable crystal plasticity and GPU execution in a recent mechanics application [15].

Bridge architectures ask a complementary question: how can JAX-style differentiation be combined with established finite-element or sparse-solver environments? The Firedrake–JAX bridge uses tangent-linear and adjoint equations derived from the high-level PDE representation, avoiding differentiation through low-level solver iterations [16]. The `FEniCSx` external-operator work of Latyshev et al. [17] uses algorithmic automatic differentiation for general constitutive models inside `FEniCSx`, including a Mohr–Coulomb example. The JetSCI preprint is especially close architecturally because it uses JAX for local finite-element discretization and PETSc for robust distributed sparse solves in heterogeneous micromechanics [18]. These works motivate the primary JAX+PETSc realization studied here. Our emphasis, relative to the cited systems, is the comparison of element AD, constitutive AD, and colored sparse recovery from AD-based Hessian-vector-product (AD-HVP) or finite-difference gradient probes on representative cases inside one nonlinear energy-minimization suite spanning the benchmark families in Section 5.

Mechanics and topology benchmark context. The plasticity benchmarks use Mohr–Coulomb-type slope-stability problems, so the relevant literature separates continuum plasticity, strength-reduction practice, constitutive integration, and endpoint comparisons. The Davis-style reduction context used in this paper traces to the plasticity treatment of Davis [26] and to later finite-element strength-reduction studies such as Tschuchnigg et al. [7]. For constitutive integration, the incremental reference frame is return mapping with consistent tangents [27, 28]. The Sysala slope-stability line provides Mohr–Coulomb source-family context, variational and optimization viewpoints for shear strength reduction [2–4], and a 3D continuation/iterative-solver reference line [5]. The Čermák–Sysala–Valdman MATLAB implementation [6] documents practical 2D/3D elastoplastic finite-element algorithms for this problem class. We use these works to position the benchmark family; the numerical claims remain limited to the implemented endpoint surrogate and the reported comparison observables.

For topology optimization, the classical solid isotropic material with penalization (SIMP) and educational references supply the compliance, filtering, and continuation ideas used throughout the field [19–22]. A relevant modern software reference is `FEniTop`, which documents a 2D/3D `FEniCSx` topology-optimization implementation with parallel support [23]. It does not duplicate the reduced design objective used here, but it contextualizes compact parallel topology optimization in the modern `FEniCSx` ecosystem.

Sparse derivative recovery and scalable nonlinear infrastructure. When exact second-order information is too expensive, sparse derivative recovery via graph coloring remains the classical alternative. Curtis, Powell, and Reid gave the foundational sparse-Jacobian viewpoint [29], while Coleman and Moré extended the approach to sparse Jacobian and Hessian estimation with graph-coloring structure [30, 31]. PETSc supplies the distributed vectors, matrices, Krylov methods, nonlinear solvers, and multigrid infrastructure needed to turn those derivative objects into scalable distributed algorithms [32, 33]. In the present toolset, PETSc ownership layout, Krylov policy, nonlinear globalization, and preconditioner design are part of the numerical method.

Against that background, the present toolset uses JAX for local element and constitutive differentiation, while PETSc owns sparse distributed algebra and nonlinear solver policy for the benchmark suite in Section 5.

3 Methodology

We use a common finite-element notation across all benchmark families. The continuous domain is $\Omega \subset \mathbb{R}^d$, and $V_{h,0} = \mathrm{span}\{\varphi_i\}_{i=1}^n$ is the conforming space of free basis functions after homogeneous essential constraints have been applied. We write $\mathbf{n}$ for the outward unit normal on boundary portions. Let θ collect benchmark parameters such as load, material, continuation, strength-reduction, and lift data. Nonhomogeneous essential data are represented by an affine lift $\bar{u}_h(\theta)$, with $\bar{u}_h = 0$ in the homogeneous cases. For vector-valued mechanics problems, $V_{h,0}$ denotes the constrained vector space, basis functions may be vector-valued, and n counts free scalar degrees of freedom. A free coefficient vector $x \in \mathbb{R}^n$ defines the finite-element state

$$u_h(x;\theta) = \bar{u}_h(\theta) + \sum_{i=1}^{n} x_i \varphi_i.$$

When no confusion is possible, we identify u_h with its coefficient vector in algebraic formulas. On element e, $x_e = R_e x$ denotes the local coefficient vector selected by the element restriction R_e, while $u_e(x;\theta)$ denotes the corresponding local element state including any contribution from the lift.

The benchmark families use two scalar functional roles. A load-step potential or endpoint surrogate is written on a mesh $\mathcal{T}_h = \{e\}_{e=1}^{n_{\mathrm{el}}}$ as

$$\Pi_h(x;\theta) = \sum_{e=1}^{n_{\mathrm{el}}} \Phi_e(x_e;\theta_e) - f_{\mathrm{ext}}^{\top} x, \tag{1}$$

where θ_e is the element-local restriction of those data and f_{ext} is the assembled load vector on the free degrees of freedom. The expression is the reduced algebraic potential in the free variables; additive constants depending only on the affine lift or prescribed loads are omitted because they do not affect the stationarity equations. A true reduced optimization objective with design variable z is written schematically as

$$\mathcal{J}_h(z) = \mathcal{R}_h(z) + \mathcal{S}_h(z, x_h(z); \theta(z)), \tag{2}$$

where $x_h(z)$ is the state determined by the benchmark state equation or stationarity condition, $\mathcal{R}_h$ contains regularization or design penalties, and $\mathcal{S}_h$ denotes the state-dependent scalar part. Some design updates instead solve frozen-state subproblems. For those steps we write

$$\mathcal{J}_h^m(z; x_h^m) = \mathcal{R}_h^m(z) + \mathcal{S}_h^m(z, x_h^m; \theta^m), \tag{3}$$

where x_h^m is fixed from the current outer iterate. The topology objective in Section 5.6 has this second form; its frozen compliance term is not the load-step potential (1). Thus (1) covers scalar PDE, mechanics, and endpoint-surrogate solves, while (2) and (3) record the two reduced-objective roles. When an algorithm is stated independently of a benchmark family, we denote its merit functional by $\mathcal{F}$. Hyperelasticity is the only family in which $J = \det F$ denotes the deformation Jacobian, so we avoid the bare symbol J for generic objectives elsewhere in the paper.

In (1), Φ_e is the integrated scalar contribution of element e. In quadrature-based mechanics examples, $\Phi_e(x_e;\theta_e) = \sum_q w_q \mathcal{W}_q(\varepsilon_q;\theta_q)$, where $\varepsilon_q := B_q u_e(x;\theta)$ is the local strain or gradient argument, q indexes quadrature points, n_q denotes the number of quadrature points on an element when finite quadrature limits are displayed, w_q includes the quadrature weight and element-map Jacobian factor, B_q is the corresponding element operator, θ_q is the quadrature-point restriction of the element data, and $\mathcal{W}_q$ is the pointwise energy density or branch potential. Within each benchmark family, all derivative routes and solver-realization comparisons use the same quadrature points and weights for the stated discrete functional; the comparisons therefore change the differentiation, assembly, or solver path, not the quadrature definition of the energy. For finite-element labels, $P_k(L_\ell)$ denotes degree-k Lagrange elements on mesh level L_ℓ; problem-specific sections define the mesh hierarchy behind those levels when it matters for interpretation.

3.1 Finite-element and derivative structure

The main design choice in the toolset is that problem definitions should remain energy-centered whenever possible. For branch potentials, all second-order objects below are active-branch derivatives away from switching sets and, for principal-value formulas, away from repeated-principal-value degeneracies. We use three derivative routes:

1. **element AD:** differentiate the full scalar element contribution $\Phi_e(x_e;\theta_e)$ with respect to the element free-DOF vector;
2. **constitutive AD:** differentiate a quadrature-point energy density $\mathcal{W}_q(\varepsilon_q, m_q)$ with respect to ε_q, holding fixed the active branch and the local material and history data m_q selected from θ_q, then assemble $r_e = \sum_q w_q B_q^{\top} \sigma_q$ and $K_e = \sum_q w_q B_q^{\top} C_q B_q$, where $\sigma_q := \partial_\varepsilon \mathcal{W}_q(\varepsilon_q, m_q)$ and $C_q := \partial^2_{\varepsilon\varepsilon} \mathcal{W}_q(\varepsilon_q, m_q)$;

3. **colored sparse recovery:** recover sparse Hessian information through directionally probed gradients or Hessian-vector products (HVPs) and graph coloring.

Element AD is the most literal route and is often the cleanest mathematically. Constitutive AD preserves the same defining scalar-energy expression but changes the differentiation boundary from the full element vector to the local strain state. This is especially effective for high-order plasticity, where the full element Hessian is often much more expensive than the 6×6 constitutive tangent. The reported colored sparse-recovery configurations use AD-HVP probes; finite-difference gradient probes are the classical sparse finite-difference variant.

3.2 Newton–Krylov globalization

The nonlinear solves are based on Newton or quasi-Newton updates, but they are not all globalized in the same way. Scalar benchmarks primarily use line-search Newton. Hyperelasticity uses a trust-region method with optional post-step line search. Some benchmark configurations use a hybrid trust-region/line-search algorithm in which a reduced trust-region model provides the direction, while a bounded line search chooses the accepted step. The line-search and trust-region ingredients follow standard globalization ideas [34, Chs. 2–4]; see also Conn et al. [35]. Algorithm 1 is a schematic solver-policy template; the numerical configurations use the stopping tolerances, caps, globalization choices, and linear-solver/preconditioner choices stated in the benchmark and results tables.

Algorithm 1 Hybrid trust-region / line-search Newton

Given x_0, trust radius Δ_0, tolerances, and evaluation routines for energy, gradient, Hessian assembly or application, and linear solves
for $k = 0, 1, \dots$ **do**
    Assemble $g_k = \nabla \mathcal{F}(x_k)$
    if convergence criteria are satisfied **then**
        terminate
    end if
    Form or define the Hessian operator $H_k = \nabla^2 \mathcal{F}(x_k)$
    Approximately solve for a Newton-like direction p_k^{N} from $H_k p_k^{\mathrm{N}} = -g_k$
    if trust region is enabled **then**
        Build a small reduced model in the span of Newton and gradient directions
        Solve the reduced trust subproblem with radius Δ_k
        Optionally compare against a pure gradient trust step
        Set d_k to the selected reduced-model direction
        Restrict the line-search interval so the trial step stays in the trust ball
    else
        Set $d_k = p_k^{\mathrm{N}}$
    end if
    Select a trial step $s_k = \alpha_k d_k$ with the stated line-search rule
    Compute $\mathrm{ared}_k = \mathcal{F}(x_k) - \mathcal{F}(x_k + s_k)$ and $\mathrm{pred}_k = -g_k^\top s_k - \frac{1}{2} s_k^\top H_k s_k$
    if trust region is enabled **then**
        accept/reject using $\rho_k = \mathrm{ared}_k / \mathrm{pred}_k$ when $\mathrm{pred}_k > 0$, otherwise reject as model failure; update Δ_k
    else
        accept according to the stated merit-decrease rule
    end if
    if the trial step is accepted **then**
        Set $x_{k+1} = x_k + s_k$
    else
        Retry with the updated globalization state or report a capped/failed solve
    end if
end for

Here ared_k is the achieved decrease in the merit functional and pred_k is the decrease predicted by the local quadratic or reduced trust-region model. A nonpositive predicted decrease is treated as model failure for the trust-region acceptance test. The admissible line-search families are Armijo backtracking, bounded merit search, and residual-bisection safeguards. An Armijo step satisfies

$$\mathcal{F}(x_k + \alpha d_k) \le \mathcal{F}(x_k) + c_1 \alpha g_k^\top d_k,$$

with $c_1 \in (0, 1)$; bounded merit search examines a prescribed bounded interval and accepts only a trial value satisfying $\mathcal{F}(x_k + \alpha d_k) < \mathcal{F}(x_k)$; residual-bisection safeguards bisect the trial step until the residual-direction test used by the configuration is satisfied; in the PETSc Newton configurations this test is $\nabla \mathcal{F}(x_k + \alpha d_k)^\top d_k < 0$, unless the line-search cap is reached. Benchmark tables instantiate the finite set of remaining policy parameters, including accept/reject thresholds, radius updates, caps, and fallback rules, wherever they affect interpretation. Here KSP denotes PETSc's Krylov-solver interface and residual tolerance, PMG denotes same-mesh polynomial multigrid, MUMPS denotes a multifrontal sparse direct solver used in coarse solves, and Hypre denotes the algebraic-multigrid/preconditioning backend used in the stated auxiliary configurations. Reported PETSc linear solves use the relative KSP residual tolerance summarized in Table 15 or stated with the corresponding result when the policy varies. Failed linear solves and non-descent directions are handled by the stated globalization safeguard or reported as capped solver behavior.

For the converged bottom-clamped three-dimensional plasticity scaling configuration reported in this paper, the chosen policy is simpler: Newton with Armijo backtracking and a same-mesh polynomial-multigrid $P_4 \to P_2 \to P_1$ hierarchy for the flexible-GMRES linear solve [34, Sec. 3.1]. The reported $\lambda_{\mathrm{sr}} = 1.55$ scaling profile uses a redundant LU/MUMPS coarse solve and PETSc relative KSP tolerance 1×10^{-2}; the auxiliary $\lambda_{\mathrm{sr}} = 1.0$ timing configurations use the Hypre algebraic-multigrid coarse PMG profile. Here λ_{sr} is the plasticity strength-reduction factor defined in section 5.5.

The numerical tables use a small common reporting vocabulary, summarized in Table 2. These terms are part of the experimental protocol: a completed endpoint solve, a fixed-work diagnostic, and a capped solve do not support the same type of claim.

Table 2: Solver-status and timing vocabulary used in the numerical tables.

Reported term	Definition in this paper	Interpretation
completed	All case-specific stopping tests are met before the nonlinear cap or wall-time cap.	Energy and observables are the values at the stated benchmark endpoint.
timeout / iteration cap	The wall-time limit or maximum nonlinear-iteration count is reached before the stated stopping tests are met.	Retained as solver-behavior evidence, not as validation evidence.
fixed work	A prescribed amount of work is performed, such as one Newton step, one linearization, or a fixed continuation schedule.	Used for cost, memory, or policy diagnostics rather than endpoint convergence claims.
wall time	Elapsed end-to-end time for the stated benchmark evaluation, including setup when that timing scope is specified.	Comparable only for configurations that share the same benchmark definition and timing scope.
reported solve time	Timer internal to the nonlinear or PETSc solve; setup, state export, and postprocessing may be outside this timer.	Used for within-table solver comparisons, not as a universal replacement for wall time.
relative correction	Norm of the accepted Newton correction relative to the current iterate scale; paired gradient checks use the absolute or relative target stated with the result.	Defines the stopping metric in 3D plasticity configurations that report correction or gradient targets.

3.3 Colored sparse recovery

Sparse derivative recovery is built around the observation that if columns that do not interfere in the sparsity pattern are perturbed together, one gradient or HVP evaluation can recover multiple Hessian columns, equivalently multiple columns of the Jacobian of the gradient, [29, 31]. The reported colored sparse-recovery path uses distance-2 coloring on the Hessian sparsity graph and AD-HVP probes; the same recovery formula also describes the classical finite-difference gradient-probe variant. This structure is especially effective when combined with overlap-domain assembly and owned-plus-ghost PETSc layouts.

Let $H := \nabla^2 \mathcal{F}(x)$ denote the Hessian of the merit functional at the current state, and let $\mathcal{P}(H)$ be its sparsity pattern. A distance-2 coloring partitions the column indices into groups $\mathcal{C}_c$ such that, for each row i, at most one column $j \in \mathcal{C}_c$ satisfies $(i, j) \in \mathcal{P}(H)$. For the color seed $v_c = \sum_{j \in \mathcal{C}_c} e_j$, where e_j denotes the jth Euclidean basis vector, one probe produces

$$y_c = \begin{cases} \big(\nabla \mathcal{F}(x + \delta v_c) - \nabla \mathcal{F}(x)\big)/\delta, & \text{finite-difference gradient probe,} \\ H v_c, & \text{AD Hessian-vector probe} \end{cases}. \tag{4}$$

In the finite-difference branch, $\delta > 0$ is the chosen gradient-probe step size. The structural separation then recovers $H_{ij} = (y_c)_i$ for $j \in \mathcal{C}_c$ and $(i, j) \in \mathcal{P}(H)$ in the AD-HVP branch. In the finite-difference branch it gives $H_{ij} = (y_c)_i + O(\delta)$ under local smoothness. The reported JAX+PETSc colored sparse-recovery cases use the AD Hessian-vector probe mode, so no finite-difference perturbation parameter is introduced in those configurations; the finite-difference formula states the classical recovery variant. After recovery, each rank scatters only owned sparse entries to the PETSc matrix, with overlap data used only for local probe evaluation.

Algorithm 2 Distributed colored Hessian recovery

```
Construct the owned part of the Hessian sparsity pattern
Compute a distance-2 coloring of the Hessian pattern
for each color group c do
    Build the combined probe vector v_c
    Evaluate one gradient difference or HVP for v_c
    Scatter recovered entries into owned sparse storage
end for
Assemble the owned rows of the distributed sparse matrix
```

3.4 Constitutive autodiff for 3D plasticity

The main 3D Mohr–Coulomb plasticity result in this paper uses constitutive AD. Rather than applying second-order AD to the full P_4 element vector, we define one scalar Mohr–Coulomb branch potential $\mathcal{W}_q(\varepsilon_q, m_q)$ per quadrature point, where m_q collects the active branch together with the local material and history data selected from θ_q. We differentiate this scalar potential with respect to the six engineering-strain components and assemble the element tangent from the resulting stress and constitutive tangent. This keeps the scalar-energy definition intact while reducing the cost of branchwise second-order information on high-order tetrahedra. The operator B_q, stress vector σ_q, and tangent C_q use the same Voigt engineering-shear convention, so σ_q is dual to the chosen strain vector.

Algorithm 3 Constitutive AD assembly for high-order plasticity

```
for each local element e do
    for each quadrature point q do
        Evaluate strain ε_q = B_q u_e(x; θ)
        Evaluate scalar energy density W_q(ε_q, m_q)
        Obtain stress σ_q = ∂_ε W_q(ε_q, m_q)
        Obtain tangent C_q = ∂²_εε W_q(ε_q, m_q)
    end for
    Assemble r_e = Σ_q w_q B_q^T σ_q
    Assemble K_e = Σ_q w_q B_q^T C_q B_q
end for
```

The key methodological point is that automatic differentiation is applied directly to the scalar branch potential defined by the benchmark. The constitutive tangent is therefore obtained from that same scalar potential rather than replaced by a separate hand-derived formula. Away from branch switching sets and repeated-principal-value degeneracies, this yields branchwise exact derivatives of the surrogate; on those nonsmooth sets the resulting tangents are branchwise surrogate derivatives rather than globally smooth continuum derivatives.

4 Implementation

Given a discrete energy density, branch potential, or constitutive law together with finite-element restrictions, the JAX+PETSc realization evaluates local residual, tangent, or Hessian-vector-product contributions with JAX. The distributed layer scatters owned coefficients to ghosted element states, accumulates owned-row sparse residuals and tangents, and gives PETSc the resulting nonlinear residual, matrix, or matrix-free action. PETSc then controls globalization, Krylov iteration, preconditioning, and convergence reporting. The same local-to-global construction is used before narrower reference formulations are introduced for selected benchmark checks.

Primary JAX+PETSc realization. This realization combines JAX-based local physics with PETSc distributed vectors, matrices, Krylov solvers, and preconditioners. Problem-specific local evaluations stay in JAX; sparse distributed assembly, nonlinear globalization, and linear algebra stay in PETSc.

Pure JAX reference implementations. These are compact differentiable realizations of selected benchmark energies. They are useful for serial formulation checks and AD diagnostics where a matched serial formulation exists, and for compact design demonstrations where the benchmark role is narrower. They are not intended to be the primary distributed route for the largest distributed benchmarks. They show how selected energy formulations can also be expressed without sparse distributed infrastructure.

FEniCS reference implementations. The FEniCS implementations remain valuable on families where they exist, especially p-Laplace, Ginzburg–Landau, and hyperelasticity. They provide a trusted high-level FEM reference and help distinguish modeling issues from solver-policy issues. In this paper they are used mainly as scoped reference comparisons rather than as the primary distributed realization.

4.1 Autodiff modes

The JAX+PETSc realization exposes several second-order modes. The precise available choices depend on the benchmark family, but the common logic is:

- **element AD**: local Hessians obtained by differentiating the full element energy on the active smooth branch;
- **constitutive AD**: branchwise local constitutive tangents assembled from quadrature-point energy densities;
- **colored sparse recovery**: structured sparse recovery from AD-HVP or finite-difference gradient probes when full element Hessians are not the appropriate cost tradeoff.

This split is scientifically important. It permits comparison of AD-derived local second-order information with carefully structured sparse recovery routes without changing the nonlinear solver or sparse assembly path. Only the finite-difference probe variant is an approximation to the directional derivative. In other words, the paper compares derivative strategies inside one common distributed solver construction, not only across reference implementations.

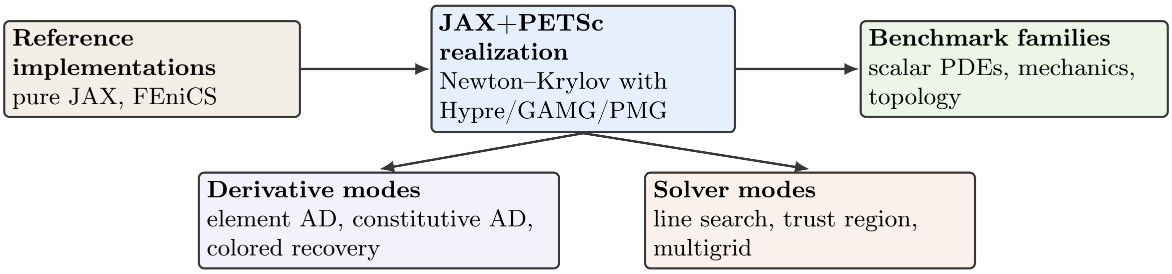


Figure 1: Computational structure used in the study. The JAX+PETSc formulation is the distributed realization; pure JAX and FEniCS provide reference formulations where matched benchmark definitions are available.

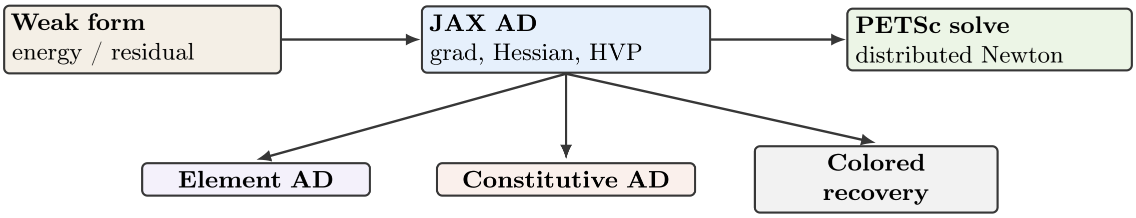


Figure 2: Derivative pathways from scalar energies and quadrature-point potentials to distributed PETSc nonlinear solves.

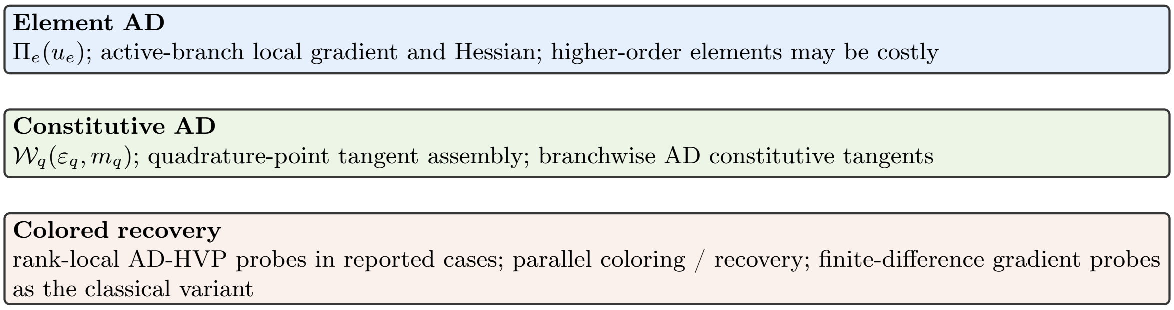


Figure 3: Second-order derivative modes exposed in the JAX+PETSc realization. The methodological choice is that the scalar energy or energy-density expression remains the defining mathematical object even when the tangent route changes from element AD to constitutive AD or colored sparse recovery.

Figures 1 to 3 summarize the computational construction used in the rest of the paper. The primary JAX+PETSc realization is the distributed nonlinear solve. Where matched definitions are available, reference formulations provide scoped comparisons; otherwise they serve as compact demonstrations. The derivative modes differ only in where second-order information is constructed before PETSc receives sparse algebraic objects.

4.2 Linear solvers and preconditioners

The benchmark families use different Krylov and preconditioner combinations because the Hessian and tangent systems have different definiteness, block structure, and coarse-level behavior. Here CG denotes conjugate gradients, GMRES generalized minimal residual, FGMRES flexible GMRES, GAMG PETSc's geometric-algebraic multigrid preconditioner, Hypre an algebraic multigrid/preconditioning backend, MUMPS a multifrontal sparse direct solver, and PMG polynomial multigrid:

- **p-Laplace:** the scalar positive-definite structure makes Newton–CG with algebraic multigrid a natural reference point for comparing exact element Hessians and colored sparse-recovery Hessians.
- **Ginzburg–Landau:** the double-well energy produces more indefinite linearizations, so GMRES with algebraic multigrid is used to separate globalization behavior from symmetric-positive-definite assumptions.
- **Hyperelasticity:** large-deformation mechanics requires globalization around the logarithmic energy domain, while the linearized systems motivate GAMG and same-hierarchy PMG variants with robust coarse solves.
- **Plasticity:** nonsmooth branch structure and high-order element spaces motivate flexible Krylov methods and same-mesh polynomial multigrid hierarchies, with direct or algebraic coarse solves used to expose the cost of the active constitutive tangent.

- **Topology:** the mechanics subproblem is solved with flexible Krylov iteration and GAMG inside an outer distributed design update, so the solver evidence reflects the coupled state-design optimization problem.

The result sections then instantiate these solver choices in concrete benchmark configurations. Some configurations deliberately hold the preconditioning operator fixed along the nonlinear path or vary only the coarse solve. These diagnostics are included to separate derivative-route cost, preconditioner construction, and outer globalization effects.

4.3 Distributed assembly and coloring

The JAX+PETSc implementation uses PETSc-owned ranges, local-plus-ghost layouts, and owned-row sparse assembly. Let I_r be the free-DOF indices owned by MPI rank r, with the sets $\{I_r\}$ forming a disjoint partition of the global free indices. Let G_r be the ghost indices needed by elements that touch I_r, and let $\widetilde{I}_r = I_r \cup G_r$. Before energy, gradient, Hessian, or HVP work, owned coefficients are scattered to $\widetilde{I}_r$; the JAX evaluations then compute element and quadrature contributions rank-locally on this overlap state.

Matrix entries are inserted only for rows in I_r, with values on G_r used to evaluate local stencils but not to duplicate ownership. Scalar energy values in the numerical tables are global reductions for the corresponding solver configuration; the ownership rule here concerns derivative and matrix assembly. Here $\mathcal{G}_i(u)$ and $\mathcal{A}_{ij}(u)$ are assembled residual and tangent entries, while $r_{e,i}$ and $a_{e,ij}$ are element contributions. Equivalently, if $\mathcal{T}_r^{\text{loc}} = \{e : \text{dofs}(e) \cap I_r \neq \emptyset\}$ and R_e is the element restriction from the ghosted rank state to element e, then $u_e = R_e u_{\widetilde{I}_r}$ and rank r inserts

$$\mathcal{G}_i(u) = \sum_{e \in \mathcal{T}_r^{\text{loc}}:\, i \in \text{dofs}(e)} r_{e,i}(u_e), \qquad \mathcal{A}_{ij}(u) = \sum_{e \in \mathcal{T}_r^{\text{loc}}:\, i,j \in \text{dofs}(e)} a_{e,ij}(u_e), \quad i \in I_r,\ j \in \widetilde{I}_r,$$

and uses ghost columns only through the local stencil. This communication order matters for both direct second-order assembly and colored sparse recovery. Exact element or constitutive assembly can write directly into local sparse buffers. Colored sparse recovery requires an additional graph-coloring step, but once the coloring is known the AD-HVP or finite-difference probe evaluations use the same ghost fill on $\widetilde{I}_r$ and still produce owned-row sparse insertions.

Line search
Newton direction; 1D merit decrease; accept by energy / residual drop

Trust region
reduced model step; ρ-based acceptance; radius shrink / expand

Hybrid
trust-model direction; bounded line search; fallback to gradient-safe step

Figure 4: Globalization choices used across the benchmark suite, from pure line search to trust-region and hybrid strategies.

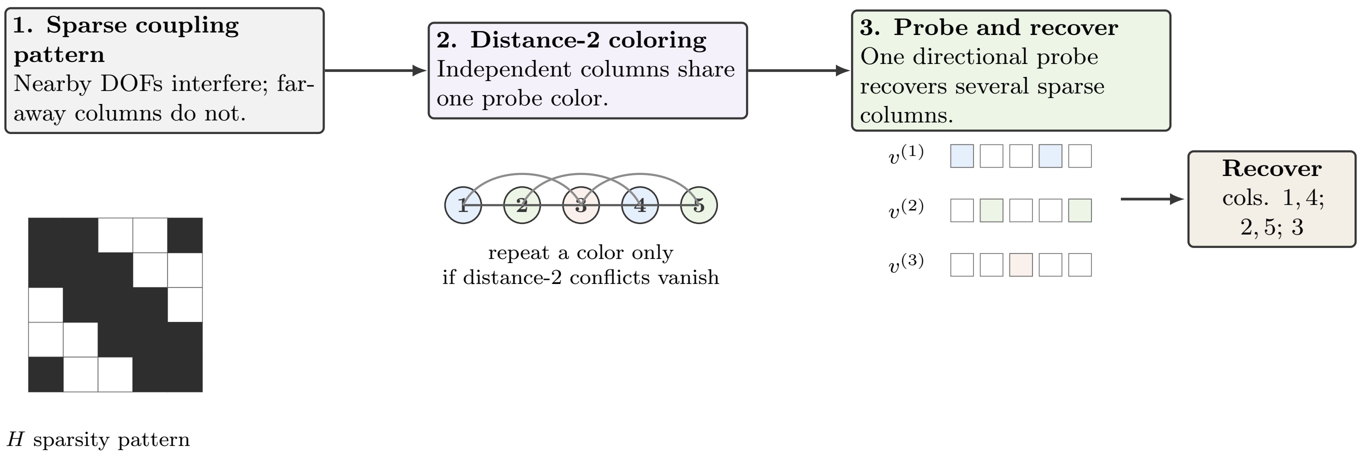


Figure 5: Schematic view of the colored sparse-recovery logic used by the colored Hessian-recovery path: a sparsity pattern, a distance-2 coloring, and the corresponding grouped probes used for recovery.

Figures 4 and 5 make two solver-policy choices explicit. Globalization is varied independently of the element physics where the benchmark permits it, while colored sparse recovery uses graph structure to batch directional Hessian probes without changing the nonlinear problem definition.

Table 3: Solver and derivative components by benchmark family. "Solver and derivative components" summarizes the derivative routes, nonlinear policies, or coupled solver features that are scientifically relevant in each family.

Family	FEniCS	pure JAX	JAX+PETSc	Solver and derivative roles
p-Laplace	yes	yes	yes	element AD and colored sparse recovery; Newton–CG with Hypre
Ginzburg–Landau	yes	no	yes	element AD and colored sparse recovery; Armijo Newton with GMRES–Hypre
Hyperelasticity	yes	yes	yes	element AD in the primary route, scoped colored sparse-recovery comparison, and trust-region/GAMG or PMG diagnostics
2D Mohr–Coulomb	no	no	yes	endpoint branch-potential derivatives and continuation diagnostics with a same-mesh PMG hierarchy
3D Mohr–Coulomb	no	no	yes	constitutive AD, element AD, and colored sparse-recovery diagnostics; FGMRES with same-mesh PMG and Hypre or LU/MUMPS coarse profiles
Topology optimization	no	yes	yes	distributed design updates with PETSc mechanics and GAMG-preconditioned FGMRES

Table 3 summarizes the solver components used in the benchmark suite. The most important structural point is that the JAX+PETSc column is the distributed algebraic realization, even when a family also has FEniCS or pure JAX reference formulations. The table is intentionally a component map rather than a full parameter schedule; configuration-specific stopping tolerances, iteration caps, timing scopes, and coarse-solver variants are reported with the numerical evidence they qualify.

5 Benchmark Problems

The benchmark suite is intentionally heterogeneous. The central claim of the paper is not that one solver policy dominates every nonlinear PDE family, but that one differentiable sparse-solver toolset can be exercised across representative scalar elliptic problems, phase-field-type energies, finite-strain mechanics, elastoplastic slope stability, and reduced topology optimization through a common energy and reduced-objective viewpoint.

Table 4: Benchmark families, reported scopes, solver policies, and principal comparison evidence. The later numerical section revisits the same families from the timing and scaling viewpoint.

Family	Reported scope	Primary policy / diagnostics	Comparison evidence	Main difficulty
p-Laplace	L_5 serial agreement; L_9 distributed scaling; L_{10} globalization diagnostic	Newton + line search	FEniCS and JAX+PETSc on the distributed case; pure JAX in the serial agreement case	nonlinear elliptic solve with exact sparse Hessians
Ginzburg–Landau	L_5 agreement; L_9 distributed scaling; L_{10} globalization diagnostic	Newton + line search	FEniCS and JAX+PETSc comparison	indefinite local curvature from the double well
Hyperelasticity	L_1 serial agreement; L_4, 24-step distributed scaling; L_5 PMG/memory diagnostics	trust-region solve	FEniCS and JAX+PETSc on the distributed suite; pure JAX only as a serial formulation check	nonconvex large-deformation mechanics
2D Mohr–Coulomb	$P_4(L_5)$–$P_4(L_7)$	endpoint solve and capped fixed work	JAX+PETSc endpoint and solver-policy evidence	same-mesh PMG and nonlinear tail behavior
3D Mohr–Coulomb	$P_1/P_2/P_4$	endpoint, scaling, and PMG-policy studies	constitutive AD, reference-formula assembly diagnostic, and PMG-policy evidence	heterogeneous 3D Mohr–Coulomb with constitutive AD
Topology	192×96 serial demonstration; 768×384 parallel benchmark; 384×192 controlled rank check	adaptive continuation	pure JAX on the serial demonstration; JAX+PETSc on the fine-grid adaptive MPI timing study and controlled rank-consistency check	distributed design-mechanics coupling

Table 5: Availability of FEniCS and pure JAX references. Reference implementations are used where documented implementations exist for the same benchmark family; absent cells are left absent rather than filled with speculative baselines.

Family	FEniCS	pure JAX	Notes
p-Laplace	yes	yes	All three implementations exist; pure JAX is used in the serial parity case.
Ginzburg–Landau	yes	no	FEniCS and JAX+PETSc form the reported comparison.
Hyperelasticity	yes	yes	pure JAX is a serial formulation check only.
2D Mohr–Coulomb	no	no	The reported realization is JAX+PETSc only.
3D Mohr–Coulomb	no	no	Reference-formula constitutive assembly is used only as a supporting comparison route.
Topology	no	yes	Parallel fine-grid realization is JAX+PETSc; pure JAX remains a compact serial design demonstration.

Tables 4 and 5 define the comparison scope before the individual models are introduced. The scalar problems and hyperelasticity have FEniCS reference formulations, while pure JAX is used only where a compact serial formulation or demonstration is available.

The two plasticity families are primarily JAX+PETSc studies. The two-dimensional case supplies supporting continuation diagnostics, and the three-dimensional case uses the Čermák–Sysala–Valdman MATLAB elastoplastic implementation as implementation context [6]. The Sysala-family slope-stability literature supplies the reference-model setting [2–5]. Neither source class is used as a path-history validation baseline.

Topology optimization uses a parallel JAX+PETSc realization and a compact serial pure JAX design demonstration, but the agreement evidence for that family is the JAX+PETSc rank-consistency check. Thus some entries below are validation comparisons, whereas others are timing or solver-policy diagnostics under a stated benchmark definition. Table 6 records the compact level notation used in the model definitions and numerical tables; the labels identify discrete problem size and mesh construction, not separate physical models.

Table 6: Meaning of the mesh-level labels used in the benchmark and results tables. The counts identify the discrete problem sizes behind the compact labels; they are not additional numerical results.

Family	Level label	Construction	Representative sizes used below
p-Laplace	L_ℓ	Uniform triangular hierarchy on the L-shaped domain.	L_9: 788,481 nodes, 1,572,864 triangles, 784,385 free scalar DOFs; L_{10}: 3,149,825 nodes, 6,291,456 triangles, 3,141,633 free scalar DOFs.
Ginzburg–Landau	L_ℓ	Uniform triangular hierarchy on the square domain.	L_9: 1,050,625 nodes, 2,097,152 triangles, 1,046,529 free scalar DOFs; L_{10}: 4,198,401 nodes, 8,388,608 triangles, 4,190,209 free scalar DOFs.
Hyperelasticity	L_ℓ	Tetrahedral cantilever hierarchy with vector displacement DOFs.	L_4: 185,249 nodes, 983,040 tetrahedra, 554,013 free displacement DOFs; L_5: 1,395,009 nodes, 7,864,320 tetrahedra, 4,178,493 free displacement DOFs.
2D Mohr–Coulomb	$P_4(L_\ell)$	Degree-four same-mesh Lagrange space on the homogeneous slope hierarchy.	$P_4(L_5)$–$P_4(L_7)$ have 332,960, 1,331,520, and 5,325,440 free displacement DOFs on 20,800, 83,200, and 332,800 triangles.
3D Mohr–Coulomb	$P_k(L_\ell)$	Degree-k same-mesh Lagrange space on the base heterogeneous tetrahedral slope and one, two, or three uniform refinements.	The P_1 levels L_1–L_4 have 10,526, 79,024, 610,964, and 4,801,816 free displacement DOFs; P_2 and P_4 use the same macro tetrahedra with elevated element degree.

5.1 p-Laplace

The literature model is the standard stationary p-Laplace Dirichlet problem and its associated energy formulation [36, Ch. 2]. Let $p > 1$, let $p' = p/(p-1)$, and let $f \in L^{p'}(\Omega)$ be the source term. The unknown is a scalar field $u : \Omega \to \mathbb{R}$ and the continuous energy is

$$\mathcal{E}(u) = \int_\Omega \frac{1}{p}|\nabla u|^p \,\mathrm{d}x - \int_\Omega f u \,\mathrm{d}x, \tag{5}$$

whose Euler–Lagrange equation is

$$-\nabla \cdot \left(|\nabla u|^{p-2}\nabla u\right) = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{6}$$

The weak problem is: find $u \in W_0^{1,p}(\Omega)$ such that

$$\int_\Omega |\nabla u|^{p-2}\nabla u \cdot \nabla v \,\mathrm{d}x = \int_\Omega f v \,\mathrm{d}x \qquad \forall v \in W_0^{1,p}(\Omega). \tag{7}$$

The benchmark used in this paper takes the L-shaped domain $\Omega = (0,2)^2 \setminus [1,2]^2$, exponent $p = 3$, forcing $f = -10$, and a conforming P_1 triangular hierarchy. Here and below, $\mathcal{E}_h$ denotes the finite-element restriction of $\mathcal{E}$ evaluated with the quadrature used for the numerical comparisons. Let $V_h \subset W_0^{1,p}(\Omega)$ be the chosen conforming finite-element subspace. The discrete problem is

$$u_h \in V_h : \qquad u_h = \arg\min_{v_h \in V_h} \mathcal{E}_h(v_h). \tag{8}$$

This is the most direct benchmark in the paper for checking exact gradients, exact Hessians, and sparse-recovery alternatives against one another without constitutive or geometric complications.

Figure 6 shows the representative state on the L-shaped domain. The visible corner-driven variation is the geometric feature against which the derivative routes are compared in the energy plot and summary table.

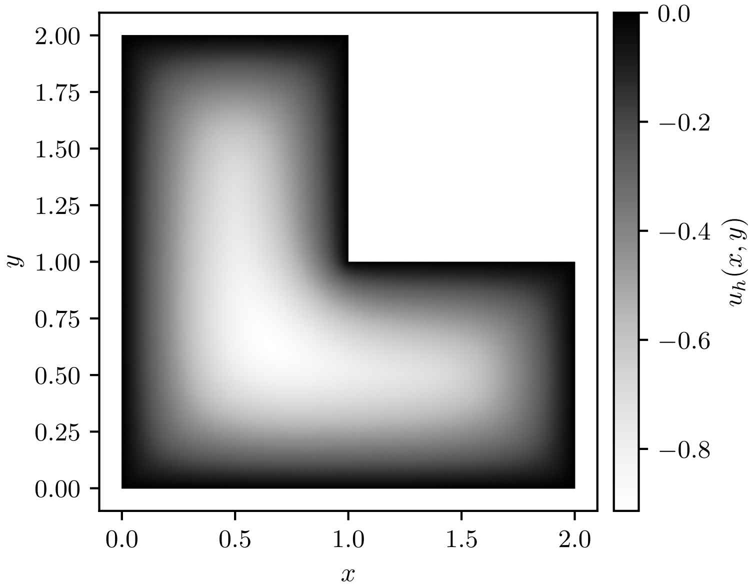


Figure 6: Representative p-Laplace state on the selected hierarchy.

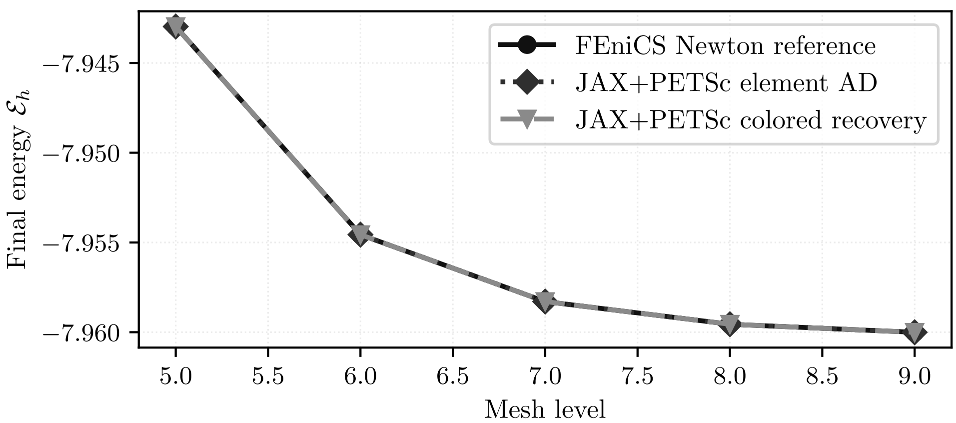


Figure 7: Final discrete energies from (8) across the selected mesh hierarchy. The element-AD and colored sparse-recovery JAX+PETSc paths are energy-consistent with the FEniCS reference at the plotted scale.

Table 7: Representative final discrete energies from (8) on the selected case.

Solver realization	Energy	Newton iters	Krylov iters	Wall time [s]
FEniCS Newton reference	−7.942968719	5	15	0.0782
pure JAX serial check	−7.943	6	0	0.1
JAX+PETSc element AD	−7.942968719	6	17	0.0798
JAX+PETSc colored recovery	−7.942968711	6	6	0.194

The within-benchmark picture is therefore one of formulation agreement: Figure 7 shows plotted-scale energy agreement across the hierarchy, and Table 7 reports the selected case where the JAX+PETSc discrete energies are within about 1×10^{-8} of the FEniCS Newton reference for (8). The pure-JAX serial entry is a compact formulation check on the same case; it is not used as distributed performance evidence. Scaling and timing for this family are reported in Section 7.3.

5.2 Ginzburg–Landau

We cite Ginzburg and Landau [37] for the historical origin of Ginzburg–Landau free-energy modeling only. The benchmark used here is a scalar real-valued double-well problem on $\Omega = [-1, 1]^2$ with homogeneous Dirichlet boundary conditions and energy

$$\mathcal{E}(u) = \int_\Omega \frac{\varepsilon}{2}|\nabla u|^2 + \frac{1}{4}(u^2 - 1)^2 \, \mathrm{d}x, \qquad \varepsilon = 10^{-2}. \tag{9}$$

Its strong form is

$$-\varepsilon \Delta u + u(u^2 - 1) = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \tag{10}$$

and the weak problem is: find $u \in H_0^1(\Omega)$ such that

$$\int_\Omega \varepsilon \nabla u \cdot \nabla v + (u^2 - 1)uv \,\mathrm{d}x = 0 \qquad \forall v \in H_0^1(\Omega). \tag{11}$$

Let $V_h \subset H_0^1(\Omega)$ be the conforming P_1 space on the uniform triangular hierarchy. The discrete stationary problem is

$$u_h \in V_h : \qquad D\mathcal{E}_h(u_h)[v_h] = 0 \qquad \forall v_h \in V_h. \tag{12}$$

Relative to p-Laplace, the additional difficulty here is not geometry but indefinite local curvature from the double-well potential in (9). All compared routes use the same initial state

$$u_0(x, y) = \sin\left(\frac{\pi(x-1)}{2}\right) \sin\left(\frac{\pi(y-1)}{2}\right),$$

restricted to the free degrees of freedom, so the selected stationary basin from this prescribed initial state is part of the benchmark definition.

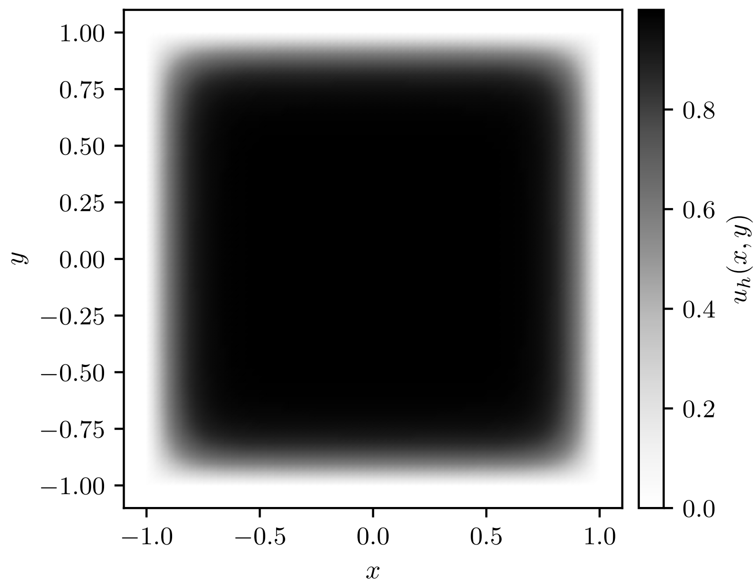


Figure 8: Representative Ginzburg–Landau state on the selected hierarchy.

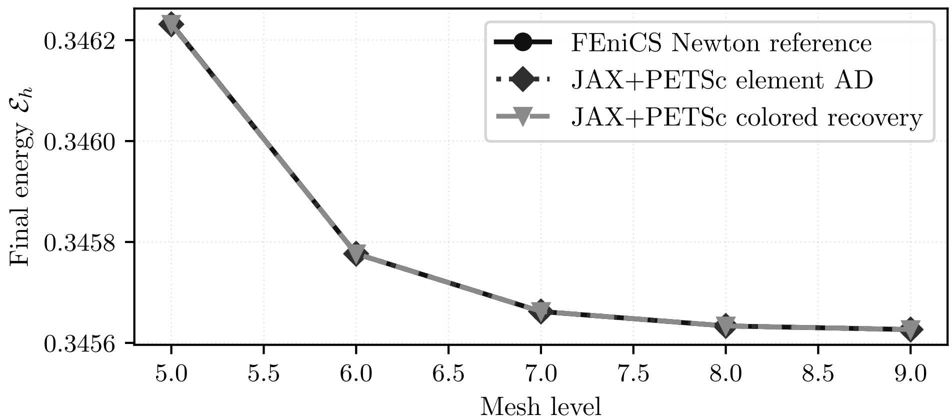


Figure 9: Final discrete energies associated with the stationary problem (12) across mesh levels. The element-AD and colored sparse-recovery JAX+PETSc routes agree with the FEniCS reference within about 1×10^{-6} in final energy.

Table 8: Representative final discrete energies associated with (12) on the selected case.

Solver realization	Energy	Newton iters	Krylov iters	Wall time [s]
FEniCS Newton reference	0.3462323651	8	33	0.113
JAX+PETSc element AD	0.3462314438	8	33	0.0949
JAX+PETSc colored recovery	0.3462314438	8	33	0.247

Figures 8 and 9 and Table 8 show endpoint-state consistency from the prescribed initial state. The benchmark evidence is again one of formulation consistency, now for a more solver-sensitive nonconvex energy: the FEniCS reference, element AD, and colored sparse recovery report matching Newton–Krylov counts and final energies within about 1×10^{-6}. Strong scaling and solver-time comparisons for this family are reported in Section 7.4.

5.3 Hyperelasticity

Hyperelasticity is the first benchmark where finite-strain mechanics and nonlinear globalization become central. The literature model is a standard finite-strain equilibrium problem written in terms of a displacement u and deformation $y_u(X) = X + u(X)$. We define the deformation gradient $F(u) = I + \nabla u$, its determinant $J(u) = \det F(u)$, and the compressible Neo-Hookean-type density

$$W(F) := C_1 \left(\operatorname{tr}(F^\top F) - 3 - 2\ln\det F\right) + D_1(\det F - 1)^2.$$

The first Piola stress is $P(F) = \partial W/\partial F$ [38, Chs. 4–6].

The benchmark considered here is a cantilever on $\Omega = [0, 0.4] \times [-0.005, 0.005]^2$ with $C_1 \approx 3.846 \times 10^7$ and $D_1 \approx 8.333 \times 10^7$. On admissible displacements with $J(u) > 0$ almost everywhere, the elastic potential is

$$\Pi(u) = \int_\Omega W(F(u))\,\mathrm{d}x, \tag{13}$$

which is the quantity compared in the benchmark summaries.

The left face is clamped, while the right face carries a prescribed rotating Dirichlet displacement path. Let $\Gamma_\mathrm{L} = \{0\} \times [-0.005, 0.005]^2$ and $\Gamma_\mathrm{R} = \{0.4\} \times [-0.005, 0.005]^2$ denote the left and right end faces, respectively. For $X = (X_1, X_2, X_3) \in \Gamma_\mathrm{R}$ and $N = 24$ load steps, the prescribed position at step k is

$$\bar{y}_k(X) = \big(X_1, \cos\eta_k\, X_2 + \sin\eta_k\, X_3, -\sin\eta_k\, X_2 + \cos\eta_k\, X_3\big), \qquad \eta_k = \frac{8\pi k}{N}.$$

Thus admissible displacements satisfy $u = 0$ on Γ_L and $X + u = \bar{y}_k(X)$ on Γ_R.

Let $V(\eta_k)$ denote the corresponding affine displacement space, let V_0 denote the homogeneous test space, and let $\Gamma_N = \partial\Omega \setminus (\Gamma_\mathrm{L} \cup \Gamma_\mathrm{R})$. The strong form is

$$-\nabla \cdot P(F(u)) = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \Gamma_\mathrm{L}, \qquad X + u = \bar{y}_k(X) \quad \text{on } \Gamma_\mathrm{R}, \qquad P(F(u))\mathbf{n} = 0 \quad \text{on } \Gamma_N, \tag{14}$$

where $P = \partial W/\partial F$ is the first Piola stress. Equivalently, the weak form is to find $u \in V(\eta_k)$ such that

$$\int_\Omega P(F(u)) : \nabla v\,\mathrm{d}x = 0 \qquad \forall v \in V_0. \tag{15}$$

Trial states with nonpositive determinant fall outside the logarithmic energy domain and are rejected by the nonlinear globalization; accepted line-search or trust-region steps are therefore evaluated only on states with positive $J(u)$.

For the discrete problem, let $V_h(\eta_k) \subset V(\eta_k)$ be the first-order tetrahedral vector finite-element space, let $V_{h,0} \subset V_0$ be its homogeneous subspace, and let Π_h be the finite-element discretization of (13). We solve the sequence of stationary points

$$u_h^{(k)} \in V_h(\eta_k): \qquad D_u\Pi_h(u_h^{(k)}; \eta_k)[v_h] = 0 \qquad \forall v_h \in V_{h,0}, \qquad k = 1, \ldots, 24, \tag{16}$$

with the boundary data above. This sequence is the basis for the hyperelastic endpoint-energy, scaling, and globalization comparisons reported below.

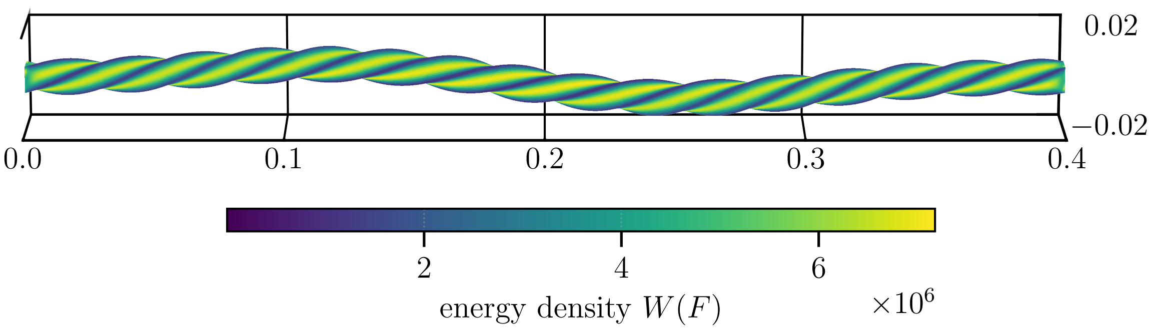


Figure 10: Representative finite-strain hyperelastic state, colored by the density W entering (13).

The within-benchmark takeaway is that all three implementations complete the 24-step boundary schedule with comparable endpoint energies. Figures 10 and 11 show the endpoint state and energy trend, and Table 9 gives endpoint energies and work counts. This is not trajectory-level agreement or a performance ranking; scaling and globalization are in Section 7.5, and the separate level-1 JAX-FEM translation companion is in Section 6.2.

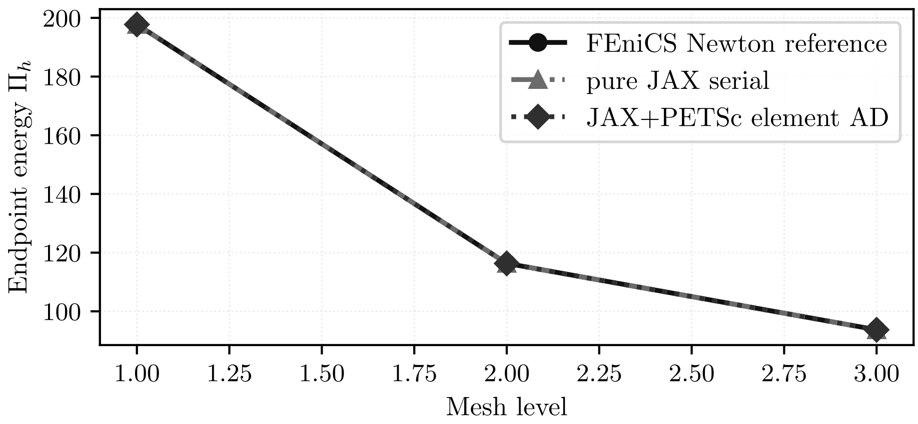


Figure 11: Final discrete energies along the stationary sequence (16) across the selected mesh hierarchy. The comparison records completion of the same 24-step schedule and comparable endpoint energies; this figure does not assess trajectory-level agreement.

Table 9: Representative 24-step endpoint energies along (16).

Solver realization	Energy	Steps	Krylov iters	Wall time [s]
FEniCS Newton reference	197.775074	24	8722	6.51
JAX+PETSc element AD	197.754582	24	10595	9.36
pure JAX serial check	197.749509	24	2284	41.5

5.4 2D Mohr–Coulomb Slope Stability

The literature model is a quasi-static plane-strain Mohr–Coulomb slope stability problem posed as equilibrium together with a constitutive update and history variables [2, 3, 7, 26–28]. The unknown is the displacement field $u = (u_x, u_y) : \Omega \to \mathbb{R}^2$ on a homogeneous slope geometry. The computational slope is the polygon with vertices $(0, 0)$, $(40, 0)$, $(40, 10)$, $(25, 10)$, $(15, 20)$, and $(0, 20)$, traversed counterclockwise. Equivalently, the lower bench width is 15, the slope run is 10, the upper bench width is 15, the two vertical height segments have length 10, and the slope angle is 45°. The bottom edge is fixed in both displacement components, the left and far-right vertical sides are fixed horizontally, and the sloping top is traction-free. Let Γ_{bot} denote the bottom edge and let Γ_{side} denote the constrained vertical side edges. We use $\Gamma_{D,x} = \Gamma_{\text{bot}} \cup \Gamma_{\text{side}}$, $\Gamma_{D,y} = \Gamma_{\text{bot}}$, and $\Gamma_N = \partial\Omega \setminus (\Gamma_{\text{bot}} \cup \Gamma_{\text{side}})$. The material parameters are $E = 40{,}000$, $\nu = 0.30$, $c_0 = 6.0$, $\phi = 45°$, $\psi = 0°$, and gravity magnitude $\gamma = 20$, so $b = (0, -\gamma)^\top$ in the endpoint solves. These values use one consistent numerical unit convention, and angles are given in degrees.

For tensor contractions in the continuum equations, write $\varepsilon^{\text{ten}}(v) = \operatorname{sym} \nabla v$. The branch potentials below use engineering Voigt strain vectors. The incremental notation in the next equations states the source continuum model; the reported discrete computations use the endpoint surrogate with zero previous plastic strain introduced later in this subsection. In the reported endpoint solves, the prescribed displacement values are homogeneous, so $\bar{u}_x^{n+1} = \bar{u}_y^{n+1} = 0$. With local load-increment index n, the continuum equilibrium equations at increment $n + 1$ are

$$-\nabla \cdot \sigma^{n+1} = b^{n+1} \quad \text{in } \Omega, \qquad u_x^{n+1} = \bar{u}_x^{n+1} \quad \text{on } \Gamma_{D,x}, \qquad u_y^{n+1} = \bar{u}_y^{n+1} \quad \text{on } \Gamma_{D,y}, \qquad \sigma^{n+1}\mathbf{n} = t^{n+1} \quad \text{on } \Gamma_N, \tag{17}$$

where σ^{n+1} and the internal variables are obtained from the elastoplastic update driven by $\varepsilon^{\text{ten}}(u^{n+1})$ and the previous state.

The endpoint studies set the old plastic strain to zero at all quadrature points. The homogeneous test space V_0 contains functions with $v_x = 0$ on $\Gamma_{D,x}$ and $v_y = 0$ on $\Gamma_{D,y}$. The corresponding weak form is

$$\int_\Omega \sigma^{n+1} : \varepsilon^{\text{ten}}(v) \, \mathrm{d}x = \int_\Omega b^{n+1} \cdot v \, \mathrm{d}x + \int_{\Gamma_N} t^{n+1} \cdot v \, \mathrm{d}s \qquad \forall v \in V_0. \tag{18}$$

The computations reported here are endpoint solves, not path-consistent incremental histories. Accordingly, we first define the local Davis-B reduction and branch potential, and then assemble them into the scalarized discrete surrogate in (20).

The two-dimensional branch formulas use the engineering Voigt strain $\varepsilon^{\text{eng}} = (\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy})^\top$. Within the local branch formulas, we write this vector as ε to keep the notation compact. The plane-strain elastic matrix is

$$C_{2\text{D}} = \begin{bmatrix} \lambda_{\text{ps}} + 2\mu & \lambda_{\text{ps}} & 0 \\ \lambda_{\text{ps}} & \lambda_{\text{ps}} + 2\mu & 0 \\ 0 & 0 & \mu \end{bmatrix}, \qquad \lambda_{\text{ps}} = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)}.$$

Its compliance is $S_{2\text{D}} = C_{2\text{D}}^{-1}$. The endpoint surrogate applies the branch return to the in-plane stress pair $(\sigma_{xx}, \sigma_{yy}, \tau_{xy})$, so principal stresses below are the two ordered in-plane values $\sigma_1^{\text{tr}} \geq \sigma_2^{\text{tr}}$.

The benchmark uses a Davis-B-style reduction path, with the Davis citation used as historical lineage and the formula context taken from later strength-reduction descriptions [3, 7, 26], to map the unreduced cohesion and angles to reduced values under strength reduction. Let $\lambda_{\mathrm{sr}} > 0$ denote the strength-reduction factor, distinct from the Lamé parameter used later in three dimensions. With c_0, ϕ, ψ denoting the unreduced cohesion, friction angle, and dilation angle, respectively, the benchmark uses

$$\begin{aligned} c_1 &= c_0/\lambda_{\mathrm{sr}}, & \phi_1 &= \arctan(\tan\phi/\lambda_{\mathrm{sr}})\,, & \psi_1 &= \arctan(\tan\psi/\lambda_{\mathrm{sr}})\,, \\ \beta &= \frac{\cos\phi_1\cos\psi_1}{1-\sin\phi_1\sin\psi_1}, & c_\lambda &= \beta c_1, & & \\ \phi_\lambda &= \arctan(\beta\tan\phi_1)\,, & \psi_\lambda &= \arctan(\beta\tan\psi_1)\,. & & \end{aligned} \tag{19}$$

The reported endpoint configurations use $\lambda_{\mathrm{sr}} = 1$ and $\psi = 0°$, hence $\psi_\lambda = 0°$; the reported states are algorithmic endpoint surrogates rather than path-consistent history updates.

The branch potential is then evaluated from the trial state. For a trial engineering strain $\varepsilon = (\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy})^\top$ and old plastic strain $\varepsilon^{p,\mathrm{old}}$, set

$$\varepsilon^e = \varepsilon - \varepsilon^{p,\mathrm{old}}, \qquad \sigma^{\mathrm{tr}} = C_{2\mathrm{D}}\varepsilon^e, \qquad S_{2\mathrm{D}} = C_{2\mathrm{D}}^{-1}.$$

Writing $\sigma^{\mathrm{tr}} = (\sigma_{xx}, \sigma_{yy}, \tau_{xy})^\top$, the principal stresses use the smoothing parameter $\epsilon_{\mathrm{reg}} = 1 \times 10^{-12}$:

$$\sigma_{1,2}^{\mathrm{tr}} = m \pm \sqrt{d^2 + \tau_{xy}^2 + \epsilon_{\mathrm{reg}}^2}, \qquad m = \frac{\sigma_{xx}+\sigma_{yy}}{2}, \qquad d = \frac{\sigma_{xx}-\sigma_{yy}}{2}.$$

With $a_\lambda = 1 + \sin\phi_\lambda$, $b_\lambda = 1 - \sin\phi_\lambda$, and $\kappa_\lambda = 2c_\lambda\cos\phi_\lambda$, the trial state is elastic when

$$f^{\mathrm{tr}} = a_\lambda\sigma_1^{\mathrm{tr}} - b_\lambda\sigma_2^{\mathrm{tr}} - \kappa_\lambda \le 0.$$

Thus the elastic branch is the branch for which the trial function in the ordered variables satisfies $f^{\mathrm{tr}} \le 0$.

In that case $\sigma^\star = \sigma^{\mathrm{tr}}$. Otherwise let

$$d_\sigma = \widehat{S}^{-1}(a_\lambda, -b_\lambda)^\top, \qquad \eta = \frac{f^{\mathrm{tr}}}{a_\lambda d_{\sigma,1} - b_\lambda d_{\sigma,2}}, \qquad \widehat{\sigma}^\ell = (\sigma_1^{\mathrm{tr}}, \sigma_2^{\mathrm{tr}})^\top - \eta d_\sigma,$$

where $\widehat{S}$ is the compliance map from the ordered principal-stress pair (σ_1, σ_2) to the corresponding normal strain pair in the trial principal basis; equivalently, it is the 2×2 normal block of $S_{2\mathrm{D}}$ after rotating from Cartesian engineering components to that basis. If $\widehat{\sigma}_1^\ell \ge \widehat{\sigma}_2^\ell$, set $\widehat{\sigma}^\star = \widehat{\sigma}^\ell$; otherwise set $\widehat{\sigma}_1^\star = \widehat{\sigma}_2^\star = \kappa_\lambda/(a_\lambda - b_\lambda)$.

Let $\vartheta = \frac{1}{2}\,\mathrm{atan2}(2\tau_{xy}, \sigma_{xx} - \sigma_{yy})$ be the trial principal direction, with the convention $\vartheta = 0$ when the trial stress has repeated in-plane principal values. This convention only fixes a representative orientation at the repeated-principal-stress branch point; the active-branch derivatives used in the solver are interpreted away from that degeneracy. We reconstruct the Cartesian stress as

$$\sigma^\star = \begin{bmatrix} \bar{\sigma}^\star + r^\star\cos(2\vartheta) \\ \bar{\sigma}^\star - r^\star\cos(2\vartheta) \\ r^\star\sin(2\vartheta) \end{bmatrix}, \qquad \bar{\sigma}^\star = \frac{\widehat{\sigma}_1^\star + \widehat{\sigma}_2^\star}{2}, \quad r^\star = \frac{\widehat{\sigma}_1^\star - \widehat{\sigma}_2^\star}{2}.$$

The branch potential is

$$\Phi_{\mathrm{MC,2D}} = \sigma^\star \cdot \varepsilon^e - \frac{1}{2}\sigma^\star \cdot S_{2\mathrm{D}}\sigma^\star.$$

Thus $\Phi_{\mathrm{MC,2D}}(\varepsilon, \varepsilon^{p,\mathrm{old}}; E, \nu, c_\lambda, \phi_\lambda)$ is the branch potential used as the two-dimensional solver-diagnostic endpoint surrogate. The small square-root regularization smooths the endpoint surrogate; it is not a separate material parameter.

Using these local definitions, the benchmark uses the following scalarized discrete surrogate functional for differentiation and comparison. Let $V_{h,0}^{2\mathrm{D}} \subset V_0$ denote the constrained discrete displacement space, and take $u_h \in V_{h,0}^{2\mathrm{D}}$. Let $c_{\lambda,eq}$ and $\phi_{\lambda,eq}$ denote the quadrature-point Davis-B-reduced cohesion and friction angle:

$$\Pi_h^{2\mathrm{D}}(u_h; \lambda_{\mathrm{sr}}) = \sum_{e=1}^{n_{\mathrm{el}}}\sum_{q=1}^{n_q} w_{eq}\,\Phi_{\mathrm{MC,2D}}\big(B_{eq}u_e, \varepsilon_{eq}^{p,\mathrm{old}}, E, \nu, c_{\lambda,eq}, \phi_{\lambda,eq}\big) - f_{\mathrm{ext}}^\top u_h, \tag{20}$$

Here the subscript eq means element e and quadrature point q, w_{eq} is the corresponding quadrature weight including the element Jacobian, and B_{eq} maps the element displacement vector to the two-dimensional engineering-strain vector used by the branch potential. The load vector f_{ext} is assembled from the right-hand side of (18); for the endpoint solves this is the gravity load with $b = (0, -\gamma)^\top$ and zero traction on the free boundary. The dilation angle enters this endpoint model only through the Davis-B reduction.

The 2D plasticity endpoint visualization reports displacement magnitude and an in-plane deviatoric engineering-strain magnitude

$$\|\varepsilon_{\mathrm{dev,2D}}\| = \sqrt{\varepsilon^\top\left(\mathrm{diag}(1,1,\tfrac{1}{2}) - \frac{1}{2}m_{2\mathrm{D}}m_{2\mathrm{D}}^\top\right)\varepsilon}, \qquad m_{2\mathrm{D}} = (1,1,0)^\top.$$

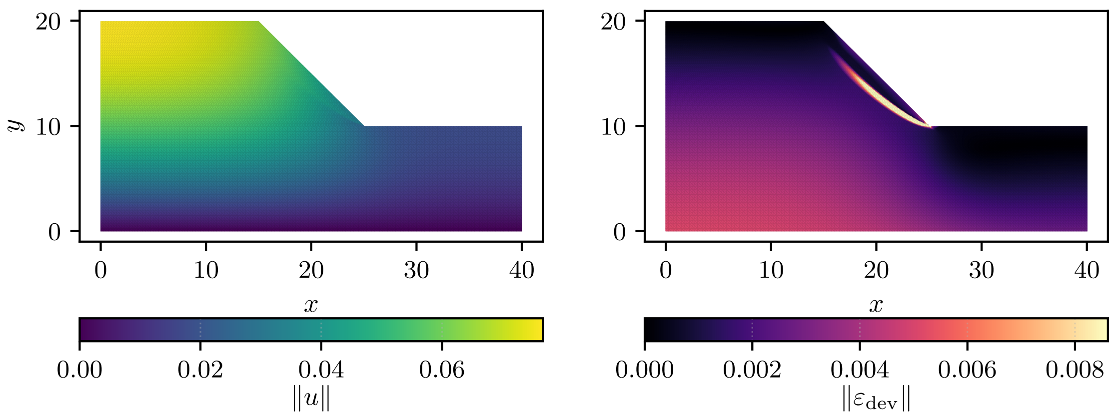


Figure 12: Representative 2D plasticity endpoint state: displacement magnitude and in-plane deviatoric strain derived from the scalar surrogate (20).

Figure 12 localizes the endpoint response in a slope mechanism region. The visualization supports the benchmark definition as an endpoint surrogate; convergence and fixed-work solver evidence are reported separately below.

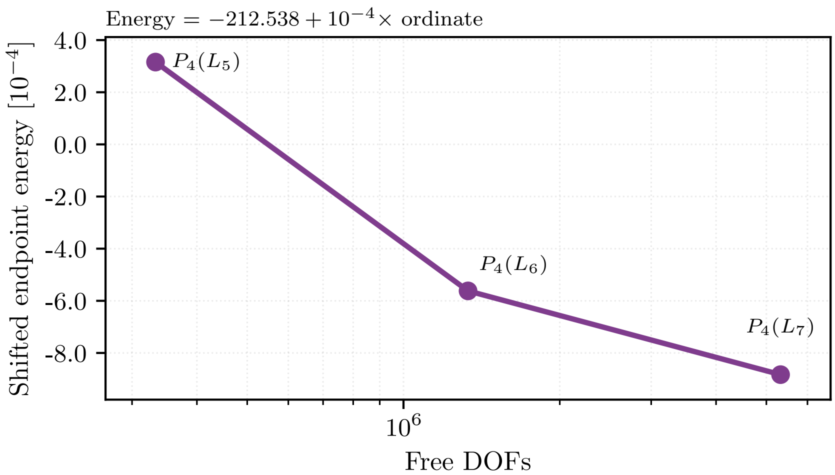


Figure 13: Resolution trend for the 2D slope-stability plasticity case. The plotted endpoint energies are evaluations of (20) on the completed $P_4(L_5)$ endpoint case and on the larger 20-iteration capped $P_4(L_6)$ and $P_4(L_7)$ diagnostics, which are not used as endpoint-convergence evidence.

Table 10: Representative 2D plasticity endpoint and fixed-work diagnostic values from (20).

Case	Free DOFs	Energy	Wall time [s]	Note
$P_4(L_5)$	332,960	−212.538	147	completed endpoint
$P_4(L_6)$	1,331,520	−212.539	147	20-iteration capped diagnostic at 8 ranks
$P_4(L_7)$	5,325,440	−212.539	253	20-iteration capped diagnostic at 16 ranks

This family is where multigrid hierarchy design first becomes dominant in the paper. Only the completed $P_4(L_5)$ endpoint case is treated as an endpoint solve; the larger $P_4(L_6)$ and $P_4(L_7)$ cases are 20-iteration capped diagnostics for solver behavior. Nonlinear and multigrid solver evidence for 2D plasticity is reported later in Sections A and 7.6.

5.5 3D Mohr–Coulomb Slope Stability

The literature model is again quasi-static Mohr–Coulomb equilibrium with constitutive/history updates, now on a three-dimensional heterogeneous slope geometry [2–5, 7, 26–28]. The unknown is a displacement field $u = (u_x, u_y, u_z) : \Omega \to \mathbb{R}^3$, and gravity acts in the vertical y-direction. Let $e_y = (0, 1, 0)^\top$. Let $\Gamma_{D,i}$ denote the component-wise Dirichlet boundary for displacement component i, let Γ_N be the remaining traction-free boundary, and let V_0 be the homogeneous vector-valued test space after applying these component-wise constraints. For tensor contractions in the continuum equations, write $\varepsilon^{\mathrm{ten}}(v) = \operatorname{sym} \nabla v$; the branch potential later uses a six-component engineering Voigt strain. The incremental notation in the next equations defines the source elastoplastic model; the reported computations use the single endpoint surrogate with zero previous history variables introduced below. For the endpoint surrogate, all prescribed displacement values on $\Gamma_{D,i}$ and on the bottom clamp are zero, so $\bar{u}_i^{n+1} = 0$. With local load-increment index n and material-region weight $\gamma(x)$, the continuum equilibrium equations are

$$-\nabla \cdot \sigma^{n+1} = -\gamma(x) e_y \quad \text{in } \Omega, \qquad u_i^{n+1} = \bar{u}_i^{n+1} \quad \text{on } \Gamma_{D,i}, \qquad \sigma^{n+1}\mathbf{n} = 0 \quad \text{on } \Gamma_N, \tag{21}$$

where σ^{n+1} and the internal variables are obtained from the elastoplastic update driven by $\varepsilon^{\text{ten}}(u^{n+1})$ and the previous state.

The heterogeneous benchmark uses a mesh-labeled material partition $\Omega = \bigcup_{m=0}^{3} \Omega_m$. The labels $m = 0, 1, 2, 3$ denote, respectively, the cover layer, general foundation, weak foundation, and slope mass. Each tetrahedron belongs to one of these regions, and all quadrature points in that tetrahedron use the corresponding tuple $(c_0, \phi, \psi, E, \nu, \gamma)$. The four tuples are, in label order,

$$\begin{aligned}
\Omega_0 &: (15, 30°, 0°, 10{,}000, 0.33, 19),\\
\Omega_1 &: (15, 38°, 0°, 50{,}000, 0.30, 22),\\
\Omega_2 &: (10, 35°, 0°, 50{,}000, 0.30, 21),\\
\Omega_3 &: (18, 32°, 0°, 20{,}000, 0.33, 20).
\end{aligned}$$

Thus, at quadrature point q, the body force is $-\gamma_q e_y$, where γ_q is selected by the material region of the containing tetrahedron. These values use one consistent numerical unit convention, and angles are given in degrees. The elastic constants used below are

$$\mu = \frac{E}{2(1+\nu)}, \qquad K = \frac{E}{3(1-2\nu)}, \qquad \lambda_{\text{L}} = K - \frac{2\mu}{3}.$$

The studied discrete model uses a heterogeneous tetrahedral mesh with four material physical groups, refined into the displayed mesh hierarchy and elevated locally to same-mesh $P_1/P_2/P_4$ spaces. Let $\Gamma_1, \ldots, \Gamma_5$ denote the boundary label sets of the reference mesh. In coordinates $X = (X_1, X_2, X_3)$, the mesh bounding box is $[-175, 30] \times [0, 60] \times [0, 86.6025]$ and X_2 is vertical. The labels $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4, \Gamma_5$ correspond, respectively, to the right face $X_1 = 30$, the left face $X_1 = -175$, the front face $X_3 = 0$, the back face $X_3 = 86.6025$, and the bottom face $X_2 = 0$; the remaining exterior labels contain the sloped free surface and the top face. We impose

$$\Gamma_{D,x} = \Gamma_1 \cup \Gamma_2, \qquad \Gamma_{D,y} = \Gamma_5, \qquad \Gamma_{D,z} = \Gamma_3 \cup \Gamma_4.$$

The bottom-clamped model additionally fixes all displacement components on $\Gamma_{\text{bot}} = \{X \in \partial\Omega : |X_2| \le 10^{-12}\}$. Equivalently, the bottom-clamped test space used below is

$$V_0^{\text{bc}} := \{v \in V_0 : v = 0 \text{ on } \Gamma_{\text{bot}}\}.$$

For the reported bottom-clamped endpoint solves, let $V_0^{\text{bc}} \subset V_0$ denote this homogeneous test space after adding the all-component bottom clamp. With these regionwise quantities, the weak form used by the endpoint solve is

$$\int_\Omega \sigma^{n+1} : \varepsilon^{\text{ten}}(v)\, \mathrm{d}x = -\int_\Omega \gamma(x) e_y \cdot v \, \mathrm{d}x \qquad \forall v \in V_0^{\text{bc}}. \tag{22}$$

We write $P_k(L_\ell)$ for polynomial degree k on mesh level ℓ and use $L_1, \ldots, L_4$ for the four displayed mesh levels.

Here and in the Davis-B reduction below, $\lambda_{\text{sr}} > 0$ denotes the strength-reduction factor. The representative figures use the bottom-clamped $P_4(L_2)$ endpoint at $\lambda_{\text{sr}} = 1.55$ and the completed nine-case study at the same strength-reduction factor. The $\lambda_{\text{sr}} = 1.0$ strong-scaling study is reported separately in Section 7.7 as auxiliary timing evidence.

We use the engineering-strain ordering

$$\varepsilon^{\text{eng}} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{yy} & \varepsilon_{zz} & \gamma_{xy} & \gamma_{yz} & \gamma_{xz} \end{bmatrix}^\top, \tag{23}$$

which is first mapped to tensor shear before principal values are formed. Within the branch-potential formulas, we write this engineering vector simply as ε. Here γ without coordinate subscripts denotes the material-region unit weight, whereas γ_{xy}, γ_{yz}, and γ_{xz} denote engineering shear components. Following the same later strength-reduction descriptions used for the two-dimensional case, the benchmark keeps the Davis-B reduction explicitly at quadrature points:

$$\begin{aligned}
c_{1,q} &= c_{0,q}/\lambda_{\text{sr}}, & \phi_{1,q} &= \arctan(\tan\phi_q/\lambda_{\text{sr}}), & \psi_{1,q} &= \arctan(\tan\psi_q/\lambda_{\text{sr}}),\\
\beta_q &= \frac{\cos\phi_{1,q}\cos\psi_{1,q}}{1 - \sin\phi_{1,q}\sin\psi_{1,q}}, & c_{\lambda,q} &= \beta_q c_{1,q}, & \phi_{\lambda,q} &= \arctan(\beta_q \tan\phi_{1,q}),
\end{aligned} \tag{24}$$

with

$$\bar{c}_{\lambda,q} = 2c_{\lambda,q}\cos\phi_{\lambda,q}, \qquad s_{\lambda,q} := \sin(\phi_{\lambda,q}). \tag{25}$$

These are the scalar constitutive parameters used by the 3D assembly path [3, 7, 26]. In this subsection, ϕ denotes friction angle and ψ denotes dilation angle. All 3D material tuples used in the reported configurations have $\psi = 0°$; the dilation angle therefore enters only through the Davis-B reduction above and not as a separate argument of $\Phi_{\text{MC,3D}}$. No separate reduced dilation-angle argument is carried in the 3D branch expressions below.

The local constitutive surrogate follows the piecewise branch structure of the matched endpoint reference formulation. At one quadrature point we suppress the index q and write $s_\lambda := s_{\lambda,q}$, $\bar{c}_\lambda := \bar{c}_{\lambda,q}$, $\mu := \mu_q$, $K := K_q$, and $\lambda_\mathrm{L} := \lambda_{\mathrm{L},q}$. Let $\varepsilon_1 \geq \varepsilon_2 \geq \varepsilon_3$ be the ordered principal strains obtained after converting engineering shear components to tensor shear. The branches represent the elastic case and the admissible plastic-return cases of the endpoint surrogate; away from branch interfaces the tests select one smooth branch for differentiation. The branch tests below use the convention that the elastic case satisfies $f_\mathrm{tr} \leq 0$. Let

$$I_1 = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}, \qquad Q(\varepsilon) = \varepsilon_{xx}^2 + \varepsilon_{yy}^2 + \varepsilon_{zz}^2 + \frac{1}{2}\left(\gamma_{xy}^2 + \gamma_{yz}^2 + \gamma_{xz}^2\right). \tag{26}$$

The elastic branch is

$$\Phi_\mathrm{el} = \frac{1}{2}\lambda_\mathrm{L} I_1^2 + \mu Q(\varepsilon), \tag{27}$$

and the branch-selection variables are

$$\begin{aligned}
&f_\mathrm{tr} = 2\mu((1+s_\lambda)\varepsilon_1 - (1-s_\lambda)\varepsilon_3) + 2\lambda_\mathrm{L} s_\lambda I_1 - \bar{c}_\lambda, \\
&r_\mathrm{sl} = \frac{\varepsilon_1 - \varepsilon_2}{1+s_\lambda}, \qquad && r_\mathrm{sr} = \frac{\varepsilon_2 - \varepsilon_3}{1-s_\lambda}, \\
&r_\mathrm{la} = \frac{\varepsilon_1 + \varepsilon_2 - 2\varepsilon_3}{3-s_\lambda}, && r_\mathrm{ra} = \frac{2\varepsilon_1 - \varepsilon_2 - \varepsilon_3}{3+s_\lambda}.
\end{aligned} \tag{28}$$

The symbols r_sl, r_sr, r_la, and r_ra denote return-test ratios; they are kept distinct from the material weight $\gamma(x)$ and the engineering shear components in (23).

The return denominators are

$$\begin{aligned}
&D_\mathrm{s} = 4\lambda_\mathrm{L} s_\lambda^2 + 4\mu(1+s_\lambda^2), \qquad && D_\mathrm{l} = 4\lambda_\mathrm{L} s_\lambda^2 + \mu(1+s_\lambda)^2 + 2\mu(1-s_\lambda)^2, \\
&D_\mathrm{r} = 4\lambda_\mathrm{L} s_\lambda^2 + 2\mu(1+s_\lambda)^2 + \mu(1-s_\lambda)^2, && D_\mathrm{a} = 4Ks_\lambda^2.
\end{aligned} \tag{29}$$

The corresponding branch multipliers are

$$\begin{aligned}
\Lambda_\mathrm{s} &= \frac{f_\mathrm{tr}}{D_\mathrm{s}}, \\
\Lambda_\mathrm{l} &= \frac{\mu((1+s_\lambda)(\varepsilon_1+\varepsilon_2) - 2(1-s_\lambda)\varepsilon_3) + 2\lambda_\mathrm{L} s_\lambda I_1 - \bar{c}_\lambda}{D_\mathrm{l}}, \\
\Lambda_\mathrm{r} &= \frac{\mu(2(1+s_\lambda)\varepsilon_1 - (1-s_\lambda)(\varepsilon_2+\varepsilon_3)) + 2\lambda_\mathrm{L} s_\lambda I_1 - \bar{c}_\lambda}{D_\mathrm{r}}, \\
\Lambda_\mathrm{a} &= \frac{2Ks_\lambda I_1 - \bar{c}_\lambda}{D_\mathrm{a}}.
\end{aligned} \tag{30}$$

The branch energies are

$$\begin{aligned}
\Phi_\mathrm{s} &= \Phi_\mathrm{el} - \frac{1}{2}D_\mathrm{s}\Lambda_\mathrm{s}^2, \\
\Phi_\mathrm{l} &= \frac{1}{2}\lambda_\mathrm{L} I_1^2 + \mu\left(\varepsilon_3^2 + \frac{1}{2}(\varepsilon_1+\varepsilon_2)^2\right) - \frac{1}{2}D_\mathrm{l}\Lambda_\mathrm{l}^2, \\
\Phi_\mathrm{r} &= \frac{1}{2}\lambda_\mathrm{L} I_1^2 + \mu\left(\varepsilon_1^2 + \frac{1}{2}(\varepsilon_2+\varepsilon_3)^2\right) - \frac{1}{2}D_\mathrm{r}\Lambda_\mathrm{r}^2, \\
\Phi_\mathrm{a} &= \frac{1}{2}KI_1^2 - \frac{1}{2}D_\mathrm{a}\Lambda_\mathrm{a}^2.
\end{aligned} \tag{31}$$

With these definitions, the scalar potential $\Phi_{\mathrm{MC,3D}}(\varepsilon; \bar{c}_\lambda, s_\lambda, \mu, K, \lambda_\mathrm{L})$ is the following piecewise function. Branch selection is elastic-first: the plastic branches are evaluated only when $f_\mathrm{tr} > 0$, and ties on branch interfaces are assigned by the first listed applicable branch:

$$\Phi_{\mathrm{MC,3D}}(\varepsilon; \bar{c}_\lambda, s_\lambda, \mu, K, \lambda_\mathrm{L}) = \begin{cases}
\Phi_\mathrm{el}, & f_\mathrm{tr} \leq 0, \\
\Phi_\mathrm{s}, & f_\mathrm{tr} > 0 \text{ and } \Lambda_\mathrm{s} \leq \min(r_\mathrm{sl}, r_\mathrm{sr}), \\
\Phi_\mathrm{l}, & f_\mathrm{tr} > 0 \text{ and } r_\mathrm{sl} < r_\mathrm{sr} \text{ and } r_\mathrm{sl} \leq \Lambda_\mathrm{l} \leq r_\mathrm{la}, \\
\Phi_\mathrm{r}, & f_\mathrm{tr} > 0 \text{ and } r_\mathrm{sr} < r_\mathrm{sl} \text{ and } r_\mathrm{sr} \leq \Lambda_\mathrm{r} \leq r_\mathrm{ra}, \\
\Phi_\mathrm{a}, & \text{otherwise}
\end{cases}. \tag{32}$$

The constitutive-AD path differentiates the smooth branches of (32) directly with JAX rather than substituting a hand-derived return-map tangent; the branch interfaces remain nonsmooth and are treated as part of the algorithmic constitutive surrogate rather than as a globally smooth continuum potential.

For the endpoint and continuation studies, these local definitions are assembled into the following scalar surrogate. The endpoint surrogate sets the previous plastic/history variables to zero and applies (32) to the total engineering strain at the current endpoint. Let $V_{h,0}^{3D} \subset V_0$ denote the constrained discrete displacement space, and take $u_h \in V_{h,0}^{3D}$. Let $\bar{c}_{\lambda,eq}$ and $s_{\lambda,eq}$ denote the quadrature-point Davis-B-reduced quantities, and set $\varepsilon_{eq}(u_e) = B_{eq}u_e$ for the six-component engineering strain at quadrature point q of element e:

$$\Pi_h^{3D}(u_h;\lambda_{sr}) = \sum_{e=1}^{n_{el}}\sum_{q=1}^{n_q} w_{eq}\,\Phi_{MC,3D}(\varepsilon_{eq}(u_e), \bar{c}_{\lambda,eq}, s_{\lambda,eq}, \mu_{eq}, K_{eq}, \lambda_{L,eq}) - f_{ext}^{\top}u_h, \tag{33}$$

where λ_{sr} is again the strength-reduction factor and λ_L is the Lamé parameter of the elastic predictor. The subscript eq means that each material parameter is evaluated at element e and quadrature point q. The load vector f_{ext} is assembled from the body force $-\gamma_q e_y$ in (22), with no Neumann traction contribution in the reported cases.

For the strain visualizations in this paper we use the deviatoric norm

$$\|\varepsilon_{dev}\| = \sqrt{\varepsilon^{\top}\left(\mathrm{diag}(1,1,1,\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}) - \frac{1}{3}mm^{\top}\right)\varepsilon}, \qquad m = (1,1,1,0,0,0)^{\top}, \tag{34}$$

which matches the engineering-shear convention of (23).

Equations (33) and (32) define an algorithmic endpoint surrogate rather than a path-consistent incremental-history functional. The paper therefore keeps 3D plasticity claims at the endpoint-surrogate level and treats seven-step strength-reduction endpoint observables separately from the fixed-λ_{sr} endpoint-formula diagnostic in Section 6.3. The related return-mapping, optimization, convex-optimization, and continuation literature [2–5] provides reference-model context for that boundary.

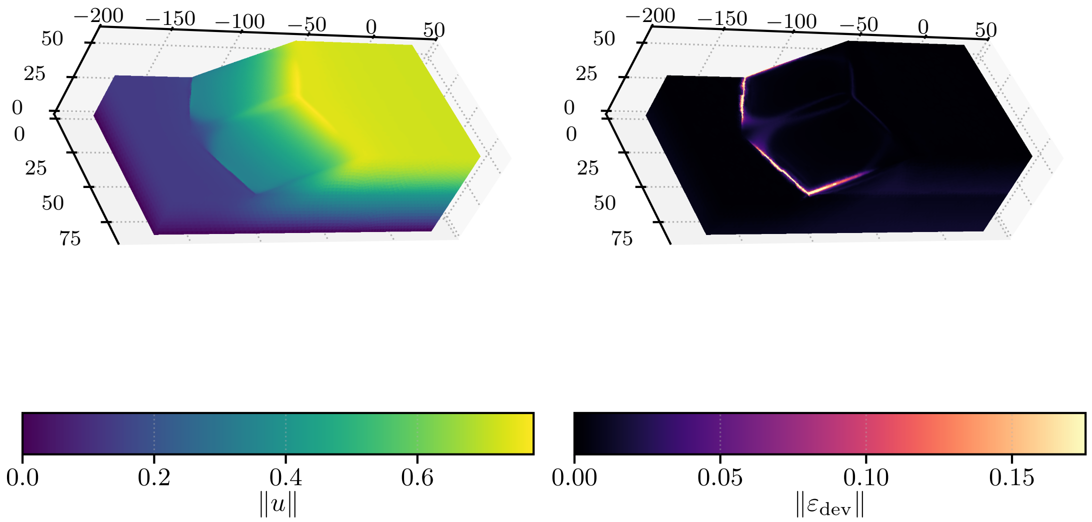


Figure 14: Bottom-clamped 3D plasticity representative endpoint at $\lambda_{sr} = 1.55$: displacement magnitude and surface deviatoric strain for the $P_4(L_2)$ configuration, both derived from the scalar surrogate in (33).

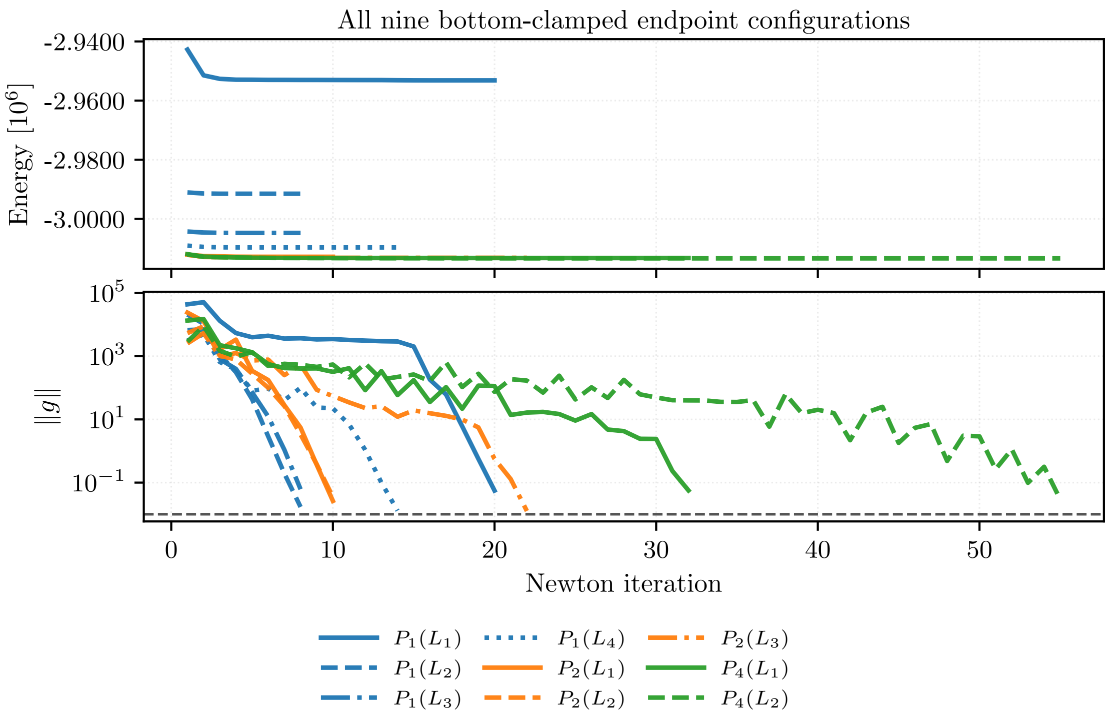


Figure 15: Newton-history convergence for all nine completed bottom-clamped 3D plasticity endpoint configurations at $\lambda_{\text{sr}} = 1.55$. The upper panel shows the value of the algorithmic surrogate from (33); the lower panel shows the gradient norm used as the nonlinear stopping metric, with color indicating degree and line style indicating mesh level.

Table 11: All nine completed bottom-clamped 3D plasticity study cases at $\lambda_{\text{sr}} = 1.55$, reported as endpoint values of (33).

Element	Free DOFs	Energy	$\|g\|_{\text{final}}/\text{target}$	Wall time [s]	Status
$P_1(L_1)$	10,526	$-2{,}953{,}167$	0.456	1.79	completed
$P_1(L_2)$	79,024	$-2{,}991{,}497$	0.164	8.72	completed
$P_1(L_3)$	610,964	$-3{,}004{,}742$	0.407	17.2	completed
$P_1(L_4)$	4,801,816	$-3{,}009{,}658$	0.116	96.7	completed
$P_2(L_1)$	79,024	$-3{,}012{,}813$	0.247	18.3	completed
$P_2(L_2)$	610,964	$-3{,}013{,}032$	0.296	98.3	completed
$P_2(L_3)$	4,801,816	$-3{,}013{,}149$	0.114	1330	completed
$P_4(L_1)$	610,964	$-3{,}013{,}227$	0.553	208	completed
$P_4(L_2)$	4,801,816	$-3{,}013{,}348$	0.442	2298	completed

Figures 14 and 15 and table 11 define the endpoint-surrogate benchmark object used later in the results section. The nine completed endpoint solves converge in the nonlinear stopping metric while the endpoint values of (33) decrease along both mesh and degree refinements.

Figure 16 adds a shared-scale slice comparison. The larger values of $\|\varepsilon_{\text{dev}}\|$ remain localized near the same upper slope region across P_1, P_2, and P_4, so the later degree and timing comparisons are interpreted as endpoint-solver comparisons rather than changes of the physical problem.

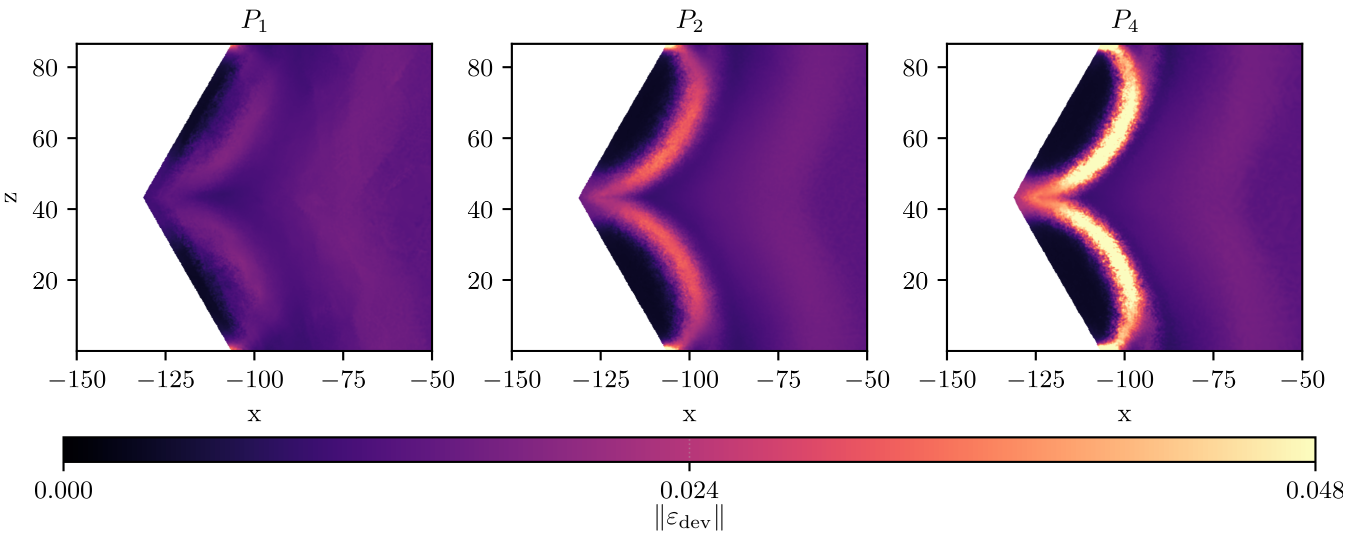


Figure 16: Shared-scale upper y-slice comparison of $\|\varepsilon_{\mathrm{dev}}\|$ from (34) at the highest mesh level of each degree line: $P_1(L_4)$, $P_2(L_3)$, and $P_4(L_2)$. The three solutions use the bottom-clamped problem and solver policy at $\lambda_{\mathrm{sr}} = 1.55$ and are compared on one common color scale.

The 3D plasticity case is the benchmark where detailed constitutive definitions are most important: the equations above define the endpoint-surrogate quantities reported in the results section, namely (33) and (34). Timing, scaling, and degree-versus-resolution evidence for this family are reported in Section 7.7.

5.6 Topology Optimization

The topology benchmark tests the toolset on a coupled design–mechanics loop rather than on a single equilibrium minimization. The literature setting is the standard SIMP-style compliance topology-optimization problem on a cantilever domain, together with filtering and continuation ideas from the classical topology literature [19–22]. In the setting studied here, each outer iteration solves a linear elasticity problem for the current density and then minimizes a frozen reduced design objective. The pure JAX topology realization is only a compact serial design demonstration; the agreement evidence for this family is the JAX+PETSc rank-consistency check in Section 7.8.

The fine-grid topology problem uses $\Omega = (0,2) \times (0,1)$ with a 768×384 structured triangular mesh, target volume fraction 0.4, $\theta_{\min} = 1 \times 10^{-6}$, Young's modulus $E = 1$, and Poisson ratio $\nu = 0.3$. The left edge is clamped, and a downward unit traction acts on the load patch $\{2\} \times [0.4, 0.6]$. With $h_x = 2/768$ and $h_y = 1/384$, fixed-solid design pads are imposed on $x \le 32h_x$ and on $x \ge 2 - 32h_x$, $y \in [0.4 - 32h_y, 0.6 + 32h_y]$.

The first ingredient is the design map. Let Z_h be the scalar continuous P_1 space on the mesh. The design variable is a latent nodal field $z_h \in Z_h$, mapped to a physical density through the logistic transform

$$\theta_h(z_h) = \theta_{\min} + (1 - \theta_{\min})\,\mathrm{sigmoid}(z_h), \qquad \mathrm{sigmoid}(z) = \frac{1}{1 + e^{-z}}. \tag{35}$$

Within this subsection, θ_h denotes the physical density field rather than the generic benchmark-parameter symbol used in Section 3. On each triangular element e, we use the element averages $\theta_e := |e|^{-1} \int_e \theta_h \,\mathrm{d}x$ and $z_e := |e|^{-1} \int_e z_h \,\mathrm{d}x$; for the linear elements used here these are vertex averages. The fixed-solid pads are included in the measured volume, and the normalized volume is

$$V(\theta_h) := \frac{1}{|\Omega|} \sum_{e=1}^{n_{\mathrm{el}}} |e|\,\theta_e.$$

Let $\mathcal{P}_h$ be the fixed-pad design nodes and let $\mathcal{D}_h$ be the remaining design nodes. At outer iteration m we write $\theta_h^m := \theta_h(z_h^m)$. During design updates, z_h is varied only on $\mathcal{D}_h$; on $\mathcal{P}_h$ the latent value is held at $z_{\mathrm{pad}} = 10$, so the corresponding physical density is held near $\theta_h = 1$. The initial field z_h^0 uses this same pad value on $\mathcal{P}_h$ and a constant value on $\mathcal{D}_h$ chosen so that $V(\theta_h^0) = 0.4$. No element is removed from the objective. Elements touching a pad enter the element sums through their fixed nodal pad values and through any remaining free design nodes. Outside the pads, the latent variable is unconstrained and (35) enforces $\theta_h \in [\theta_{\min}, 1]$.

The second ingredient is the mechanics solve for a fixed density field. Let V_h be the continuous vector P_1 displacement space and let $V_{h,0} \subset V_h$ impose the homogeneous left-edge clamp. Let C denote the plane-stress elasticity tensor and let p_{SIMP} be the active SIMP exponent. For fixed density, define

$$a_{\theta_h}(w_h, v_h) := \sum_{e=1}^{n_{\mathrm{el}}} |e|\,\theta_e^{p_{\mathrm{SIMP}}} \left(\varepsilon(w_h)|_e\right)^{\top} C \left(\varepsilon(v_h)|_e\right), \tag{36}$$

where $\varepsilon(v) = (\partial_x v_x, \partial_y v_y, \partial_y v_x + \partial_x v_y)^\top$, and

$$C = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}.$$

The load functional is

$$\ell(v) = \int_{\{2\}\times[0.4,0.6]} (0,-1)\cdot v \,\mathrm{d}s.$$

The mechanics step finds $u_h(\theta_h) \in V_{h,0}$ such that $a_{\theta_h}(u_h, v_h) = \ell(v_h)$ for all $v_h \in V_{h,0}$. The scalar compliance observable reported below is

$$\mathcal{C}_h(\theta_h) := \ell(u_h(\theta_h)) = a_{\theta_h}(u_h(\theta_h), u_h(\theta_h)).$$

It is an observable of the mechanics solve, whereas (37) is the frozen design subproblem solved at one outer iteration.

The third ingredient is the reduced design update. At outer iteration m, after solving for $u_h^m = u_h(\theta_h^m)$, we freeze the elementwise quantity

$$e_{e,m}^{\text{frozen}} = (\theta_e^m)^{2p_{\text{SIMP}}} \left(\varepsilon(u_h^m)|_e\right)^\top C \left(\varepsilon(u_h^m)|_e\right).$$

This is a reciprocal local compliance approximation. The mechanics stiffness in (36) is proportional to $\theta_e^{p_{\text{SIMP}}}$, so $e_{e,m}^{\text{frozen}}\theta_e^{-p_{\text{SIMP}}}$ matches the current local compliance scale at $\theta_e = \theta_e^m$ while penalizing lower trial densities. Let $(\nabla\theta_h)|_e$ denote the elementwise gradient of the interpolated density field used in the phase-field term, and define the double-well density

$$W_{\text{dw}}(\theta) := \theta^2(1-\theta)^2.$$

The scalar λ_V^m is the current effective volume multiplier, specified by (38) below, and α_{reg}, ℓ_{pf}, and μ_{move} are, respectively, the phase-field weight, phase-field length scale, and move regularization parameter. The frozen reduced objective for the next design vector z_h^{m+1} is

$$\mathcal{J}_h^m(z_h; u_h^m) = \sum_{e=1}^{n_{\text{el}}} |e| \left[e_{e,m}^{\text{frozen}}\theta_e^{-p_{\text{SIMP}}} + \lambda_V^m \theta_e + \alpha_{\text{reg}}\left(\frac{\ell_{\text{pf}}}{2}|(\nabla\theta_h)|_e|^2 + \frac{1}{\ell_{\text{pf}}}W_{\text{dw}}(\theta_e)\right) + \frac{\mu_{\text{move}}}{2}(z_e - z_e^m)^2 \right], \tag{37}$$

where the mechanics state u_h^m is fixed through $e_{e,m}^{\text{frozen}}$, θ_e and z_e are computed from the trial z_h, and z_e^m is the element average from the previous design iterate. The minimization of (37) is with respect to the free latent nodal values on $\mathcal{D}_h$; the fixed-pad nodal values on $\mathcal{P}_h$ remain constant during the design step.

The target volume fraction is imposed through an iteration-dependent effective multiplier. Let $V_\star = 0.4$ be the target value and initialize $\lambda_{\text{corr}}^0 = 0$. Before the mth design solve we use

$$r_V^m = V(\theta_h^m) - V_\star, \qquad \lambda_V^m = \lambda_{\text{ref}}^m + \lambda_{\text{corr}}^m + \rho_V r_V^m, \tag{38}$$

where λ_{ref}^m is the unweighted empirical $(1-V_\star)$-quantile, with linear interpolation, of the element samples

$$s_e^m = p_{\text{SIMP}}\, e_{e,m}^{\text{frozen}} / \max(\theta_e^m, \theta_{\min})^{p_{\text{SIMP}}+1},$$

taken once over the elements of $\mathcal{T}_h$ and without duplicated ghost cells in parallel, λ_{corr}^m is the carried multiplier correction, and $\rho_V = 10$. After the design update, with $r_V^{m+1} = V(\theta_h^{m+1}) - V_\star$, the carried correction is updated by

$$\lambda_{\text{corr}}^{m+1} = \lambda_{\text{corr}}^m + \beta_V \max(1, \lambda_{\text{ref}}^m) r_V^{m+1}, \qquad \beta_V = 12.$$

Here λ_V^m is the effective volume multiplier in the current design subproblem. The SIMP-style continuation in p_{SIMP} and the regularization terms control mesh-scale design features, following the role of filters in Bourdin [22]. The constants $\alpha_{\text{reg}} = 5\times10^{-3}$, $\ell_{\text{pf}} = 8\times10^{-2}$, and $\mu_{\text{move}} = 1\times10^{-2}$ set the phase-field and latent-move terms. These are algorithmic choices of this benchmark, not standard SIMP notation.

For topology, the relevant notion of "convergence" is this reduced objective history rather than a single equilibrium energy. Let $\theta_{\mathcal{D}}^{m,-}$ and $\theta_{\mathcal{D}}^{m,+}$ denote the active design-node density vectors before and after the mth design solve. The reported design-change diagnostic is $\|\theta_{\mathcal{D}}^{m,+} - \theta_{\mathcal{D}}^{m,-}\|_2 / \max(1, \|\theta_{\mathcal{D}}^{m,-}\|_2)$, and the state-change diagnostic is the same relative norm between consecutive pre-design density vectors $\theta_{\mathcal{D}}^{m,-}$ and $\theta_{\mathcal{D}}^{m-1,-}$.

The reported topology runs use at most 2000 outer iterations. Each mechanics step uses FGMRES with GAMG preconditioning, relative KSP tolerance 10^{-4}, and a cap of 100 Krylov iterations. Each design step applies gradient descent to (37), with at most 20 design iterations, golden-section line search on a bounded step interval, function tolerance 10^{-6}, gradient tolerance 10^{-3}, and line-search interval tolerance 10^{-1}. The outer stopping rule requires either the fixed-schedule cap or, after $p_{\text{SIMP}} \geq 4$, both the state-change and design-change diagnostics to be at most 10^{-6}.

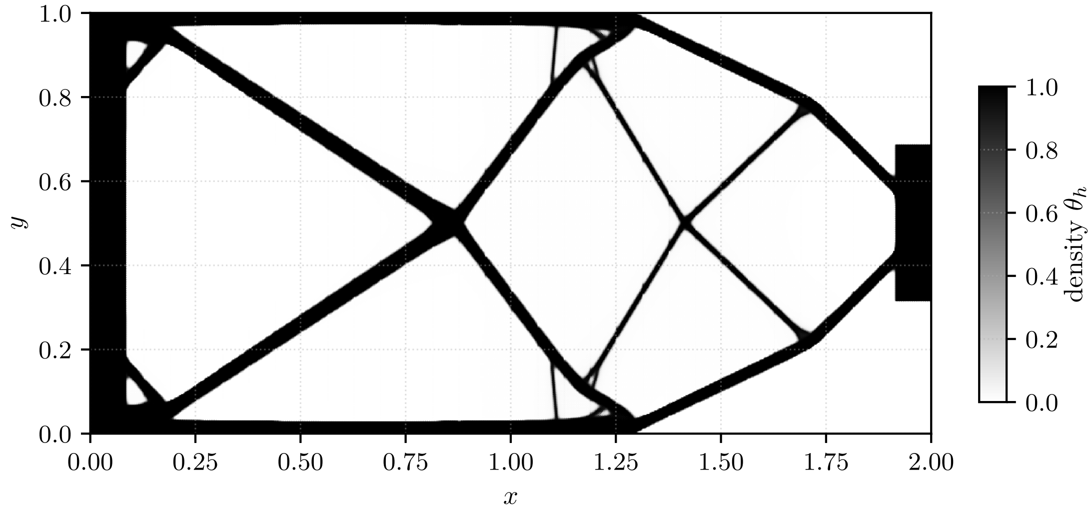


Figure 17: Representative final density field for the parallel topology benchmark.

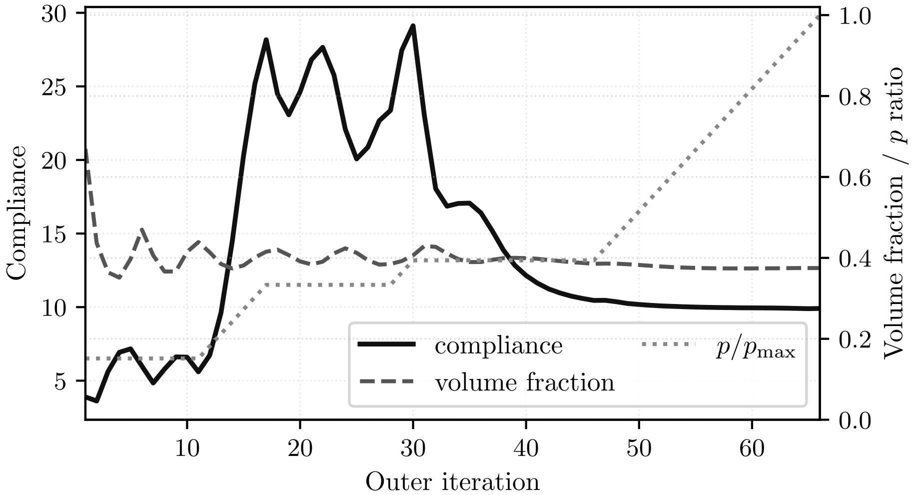


Figure 18: Objective-history view for the frozen reduced topology objective (37), with compliance on the left axis and volume fraction plus the continuation schedule in p_{SIMP} on the right axis.

Table 12: Representative topology endpoint observables for the reduced-design benchmark.

Case	Ranks	Outer iters	Compliance	Volume fraction	Wall time [s]
192×96	1	121	4.1557	0.4	10
768×384	32	66	9.9074	0.3749	25.1

Figures 17 and 18 and Table 12 show the reduced-design endpoint used for the topology timing study. The fine-grid configuration produces the displayed load-bearing density pattern, reaches volume fraction 0.3749, and stops after 66 outer iterations under the continuation logic shown in Figure 18. The target volume is imposed through the adaptive multiplier and stopping rule, so the endpoint volume is reported as an achieved value rather than as an exact equality-constraint satisfaction claim. The reported continuation starts at SIMP exponent $p_{\mathrm{SIMP}} = 1$, increases p_{SIMP} by 0.2 on each outer iteration up to $p_{\mathrm{SIMP}} = 10$, and treats the fine-grid configuration as stalled once $p_{\mathrm{SIMP}} \geq 4$ and both diagnostics are at most 10^{-6}. Adaptive timing across MPI rank counts and a controlled rank-consistency check for the same family are reported in Section 7.8.

6 Validation and External Comparisons

This section separates validation and external-comparison evidence from the performance results that follow. The distinction is important because the paper uses different levels of evidence for different claims: Hyperelasticity has a narrow external comparison against JAX-FEM, while 3D Mohr–Coulomb plasticity is evaluated through a matched endpoint-comparator formulation for the surrogate studied here.

6.1 Comparison Metrics

For vector and field comparisons with nonzero reference norm, we report relative differences of the form

$$\delta_M(a,b) = \frac{\|a-b\|_M}{\|a\|_M}, \tag{39}$$

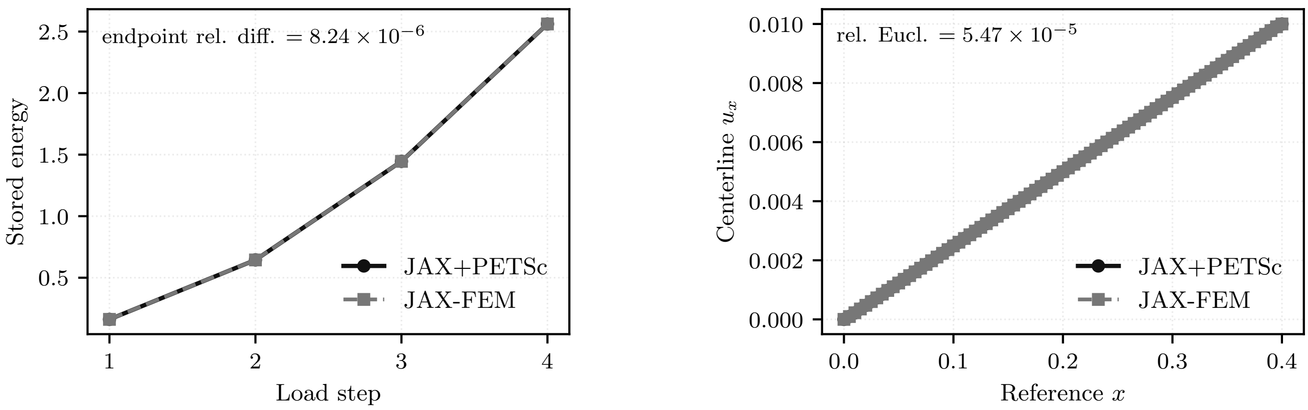
where a is the reference observable, b is the compared observable, and M denotes the discrete norm used for the comparison. For nodal displacement vectors, M is the identity on the compared free degrees of freedom. For scalar field samples r_q, we use $\|r\|_M^2 = \sum_q w_q r_q^2$. For vector-valued samples $r_q \in \mathbb{R}^m$, we use $\|r\|_M^2 = \sum_q w_q r_q^\top G r_q$, where G is stated with the observable. The reported 3D plasticity deviatoric-strain observable is the scalar field $\|\varepsilon_{\mathrm{dev}}\|$ from (34). Curve metrics are specified separately below. The reported field and profile observables have nonzero reference norms. For scalar observables, including signed energy and work, we use $|a-b|/|a|$; the hyperelastic energy post-comparison uses an absolute denominator floor of 10^{-12}. We write $u_{\max}$ for the maximum displacement magnitude over the compared state.

Curve metrics use unweighted Euclidean norms on ordered reference samples. For the hyperelastic companion, the centerline sample set is $C_{\mathrm{HE}} = \{i : \ y_i = z_i = 0\}$ in the reference coordinates $X_i = (x_i, y_i, z_i)$, ordered by x_i, and the compared observable is u_x on that set. The hyperelastic $u_{\max}$ curve uses the four prescribed right-face displacement steps as its ordered samples, and the 3D Mohr–Coulomb plasticity $u_{\max}(\lambda_{\mathrm{sr}})$ curve uses the tested strength-reduction grid; both use the same reference-denominator Euclidean norm as (39). For 3D Mohr–Coulomb plasticity, let $y_{\mathrm{mid}} = (\min_i y_i + \max_i y_i)/2$, let $L_y = \max_i y_i - \min_i y_i$, and let $q_{0.6}(z)$ be the 0.6 quantile of the reference z coordinates. The upper-slope profile sample set is $C_{\mathrm{P3D}} = \{i : \ |y_i - y_{\mathrm{mid}}| \le 0.04 L_y, \ z_i \ge q_{0.6}(z)\}$, ordered by x_i, and the compared observable is the displacement magnitude. The candidate upper-slope profile is linearly interpolated to the reference x_i coordinates before computing (39).

The hyperelastic JAX-FEM comparison uses a fixed 5% engineering gate for the final energy, Euclidean nodal full-field displacement, centerline displacement, and $u_{\max}$-curve differences. The 3D Mohr–Coulomb plasticity fixed-λ_{sr} endpoint-formula diagnostic uses three fixed criteria: at most 0.03 relative difference in the highest successful strength-reduction factor, at most 0.05 relative Euclidean difference on the ordered $u_{\max}(\lambda_{\mathrm{sr}})$ samples, and at most 0.10 relative discrete-norm difference for the endpoint displacement. The deviatoric-strain and upper-slope profile differences in Table 14 are diagnostics, not acceptance criteria. These thresholds are fixed engineering agreement gates for companion and surrogate comparisons. They are not literature validation tolerances and do not replace path-history validation for elastoplastic evolution.

6.2 Hyperelasticity External Comparison

The hyperelasticity energy is defined in Section 5.3. The external comparison uses JAX-FEM [1] in a deliberately narrow serial companion case rather than the 24-step rotating benchmark path. Both implementations use the same level-1 mesh and the same right-face x-translation schedule $d = (0.0025, 0.005, 0.0075, 0.01)$, then post-compare both endpoint states with the energy in (13). This supports endpoint-state observable agreement for this hyperelastic companion problem only. The comparison is therefore an energy-postprocessed endpoint-state check for one hyperelastic case, and it does not assert constitutive-law identity between the two implementations: the specific JAX-FEM companion case used here uses a logarithmic-J Piola form, whereas (13) uses the $D_1(J-1)^2$ volumetric penalty.



(a) Endpoint energy over matched steps.

(b) Final centerline u_x profile on the common sample path.

Figure 19: Serial hyperelastic comparison against JAX-FEM on the same mesh and prescribed displacement schedule, with energy post-comparison of the endpoint states under (13).

Table 13: Thresholded endpoint-state companion comparison against JAX-FEM, with energy post-comparison under (13).

Group	Quantity	Relative difference	Status
Agreement	final energy	8.24×10^{-6}	below 5%
Agreement	full-field displacement relative Euclidean	2.01×10^{-4}	below 5%
Agreement	centerline relative Euclidean curve norm	5.47×10^{-5}	below 5%
Agreement	$u_{\max}$ curve relative Euclidean norm	8.85×10^{-16}	below 5%
Condition	common mesh	–	satisfied
Condition	common displacement schedule	–	satisfied
Condition	agreement threshold (5%)	–	satisfied
Condition	energy re-evaluation	–	applied

Figure 19 shows that the endpoint-energy history and centerline displacement profile remain close on the matched schedule. The table quantifies that visual agreement under the common post-comparison energy.

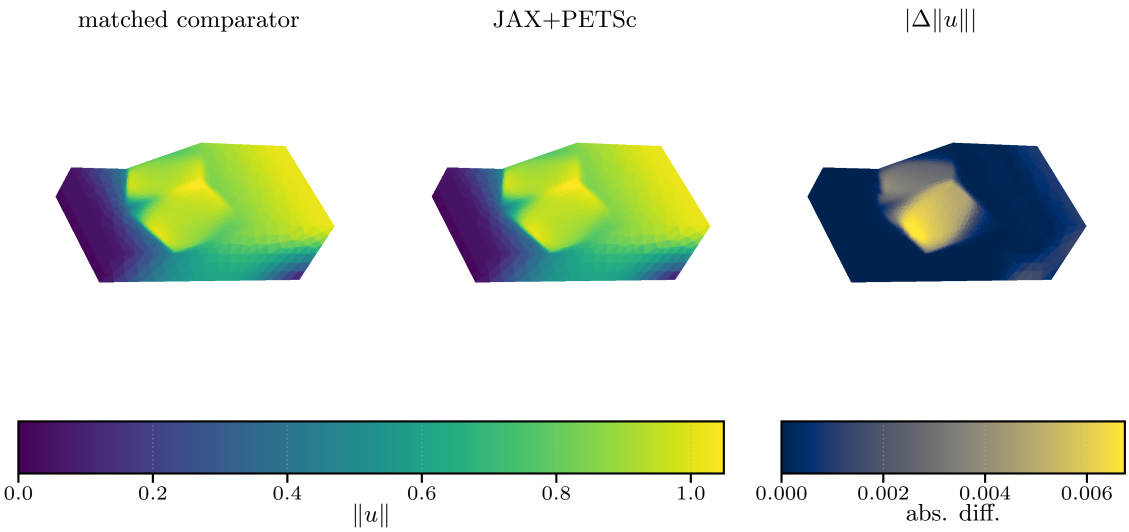
The agreement criteria in Table 13 are satisfied for all checked quantities. The final energy relative difference is 8.244×10^{-6}, the full-field displacement relative Euclidean nodal difference is 2.011×10^{-4}, the centerline relative Euclidean difference is 5.473×10^{-5}, and the $u_{\max}$-curve relative Euclidean difference is 8.85×10^{-16}. The scientific role of this table is endpoint-state agreement under a matched mesh, displacement schedule, and common energy post-comparison for this companion hyperelastic case.

6.3 3D Mohr–Coulomb Endpoint-Surrogate Comparison

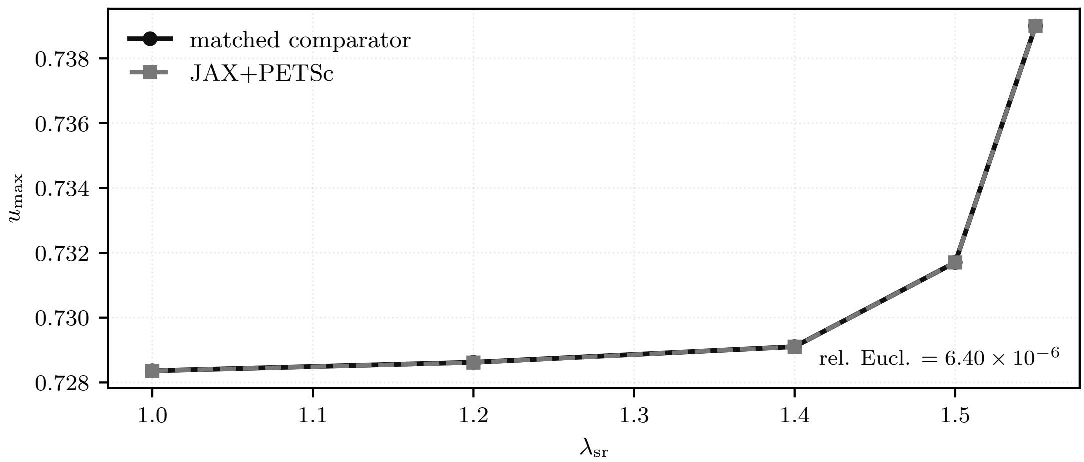
The 3D Mohr–Coulomb plasticity benchmark is tied to Mohr–Coulomb slope-stability reference formulations, but the object studied here is the algorithmic endpoint surrogate defined in Section 5.5 by (33) and (32). The Sysala return-mapping, optimization, convex-optimization, and advanced continuation papers provide reference-model and method context for this problem class [2–5]. The efficient Čermák–Sysala–Valdman MATLAB elastoplastic implementation of Čermák et al. [6] provides implementation context for practical two- and three-dimensional elastoplastic finite-element solvers. None of these sources is used here as a verified numerical baseline.

The endpoint-surrogate comparison therefore uses two diagnostics. Here the matched endpoint-formula comparator means an independent assembly of the same endpoint surrogate, not the cited incremental-history studies. The comparator uses the same endpoint functional, mesh, material table, Davis-B strength-reduction schedule, and boundary-condition convention, but evaluates the endpoint observables through that independent formula path. The endpoint-observable check tests a seven-step strength-reduction sequence, $\lambda_{\text{sr}} = 1.0, 1.1, \ldots, 1.6$, against the matched endpoint-formula comparator on the $P_2(L_1)$ comparator case. The work observable is the external work $\omega = f_{\text{ext}}^{\top} x_h$, where x_h is the free coefficient vector of the endpoint displacement u_h, reported by the two endpoint formulations. Its relative difference is 3.877×10^{-5}, the displacement relative discrete-norm difference is 3.517×10^{-3}, and the deviatoric-strain relative discrete-norm difference is 8.720×10^{-3}.

The fixed-λ_{sr} endpoint-formula diagnostic asks a narrower fixed-λ_{sr} question against the matched endpoint-formula comparator. After aligning the comparator boundary conditions with the active boundary condition set, the fixed-λ_{sr} grid is capped at the main-figure strength-reduction value $\lambda_{\text{sr}} = 1.55$. Here $\lambda_{\max}^{\text{succ}}$ denotes the largest value in the tested grid $\{1.0, 1.2, 1.4, 1.5, 1.55\}$ that satisfies the relative-correction stopping criterion. The comparison matches $\lambda_{\max}^{\text{succ}}$ on that grid and satisfies the fixed engineering criteria for the $u_{\max}(\lambda_{\text{sr}})$ curve and endpoint displacement: the $u_{\max}(\lambda_{\text{sr}})$ relative Euclidean difference is 6.396×10^{-6}, and the endpoint displacement relative discrete-norm difference is 8.299×10^{-4}. The endpoint deviatoric-strain relative discrete-norm difference (4.307×10^{-3}) and upper-slope profile relative Euclidean difference (5.229×10^{-4}) are reported as diagnostics rather than acceptance criteria. This remains an endpoint check rather than a path-consistent plastic-history reference.



(a) Endpoint-observable comparison with the matched endpoint-formula comparator.



(b) Fixed-λ_{sr} endpoint-formula diagnostic.

Figure 20: 3D plasticity endpoint-surrogate comparison. The upper panel compares the endpoint observables after the seven-step strength-reduction sequence ending at $\lambda_{\mathrm{sr}} = 1.6$; the lower panel shows the fixed-λ_{sr} endpoint-formula diagnostic after matching the boundary conditions and active free degrees of freedom.

Table 14: 3D plasticity endpoint-surrogate comparison summary for the seven-step strength-reduction endpoint-observable comparison ending at $\lambda_{\mathrm{sr}} = 1.6$ and the fixed-λ_{sr} endpoint-formula diagnostic. The gate column marks which quantities are acceptance checks and which are reported diagnostics.

Check	Observable	Rel. diff.	Gate	Status
endpoint observable	work	3.88×10^{-5}	reported only	reported
endpoint observable	displacement relative discrete norm	3.52×10^{-3}	reported only	reported
endpoint observable	deviatoric-strain relative discrete norm	8.72×10^{-3}	reported only	reported
fixed-λ_{sr} comparison	relative difference in $\lambda_{\max}^{\mathrm{succ}}$	0	≤ 0.03	satisfied
fixed-λ_{sr} comparison	$u_{\max}(\lambda_{\mathrm{sr}})$ relative Euclidean curve norm	6.4×10^{-6}	≤ 0.05	satisfied
fixed-λ_{sr} comparison	endpoint displacement relative discrete norm	8.3×10^{-4}	≤ 0.10	satisfied
fixed-λ_{sr} comparison	endpoint deviatoric-strain relative discrete norm	4.31×10^{-3}	diagnostic	diagnostic
fixed-λ_{sr} comparison	upper-slope profile relative Euclidean curve norm	5.23×10^{-4}	diagnostic	diagnostic
fixed-λ_{sr} comparison	fixed-λ_{sr} criteria, 3/3 satisfied	–	summary	satisfied

Figure 20 shows the two endpoint-surrogate diagnostics used for the comparison: the seven-step strength-reduction observable check and the fixed-$\lambda_{\rm sr}$ endpoint-formula curve after boundary-condition matching. The resulting interpretation is limited but useful. 3D Mohr–Coulomb plasticity is reported here through seven-step endpoint-observable differences, while the endpoint-formula diagnostic at fixed $\lambda_{\rm sr}$ supports matched endpoint observables for that surrogate under matched boundary conditions and active free degrees of freedom. Neither check is a path-consistent incremental-plasticity history validation.

7 Performance and Solver Behavior

Having defined the benchmark models and validation evidence, we now report computational behavior: globalization, derivative-route cost, sparse-solver policy, memory, and scaling. Timing entries from validation studies are used only when they clarify solver behavior; validation and performance interpretations remain separate. Table 15 fixes the reporting terms used across the result blocks and follows the protocol in Table 2.

Table 15: Numerical evidence blocks and reporting protocol. Validation entries fix terminology only; performance claims use the stated timing, stopping, and solver-policy columns. The KSP column gives representative tolerance and iteration policies, overridden by result-specific entries.

Evidence block	Role	Solver policy	KSP rtol / max iters	Stop or fixed work	Reported time
Scalar benchmarks	formulation, derivative-route, globalization	Newton–Armijo or stated trust-region variant; PETSc entries use CG/GMRES with Hypre	p-Laplace: $10^{-1}/30$; Ginzburg–Landau: $10^{-3}/200$	case-specific nonlinear tolerances; capped configurations diagnostic	wall or solve/elapsed time
Hyperelasticity	finite-strain mechanics, JAX-FEM comparison, scaling	trust-region or line-search Newton; GAMG or PMG as stated	GAMG entries: $10^{-1}/30$; PMG tolerances stated in diagnostic entries	per-load stationarity; fixed-step entries diagnostic	wall time; reported solve time for first-step scaling
2D Mohr–Coulomb	endpoint evidence and fixed-work diagnostics	Armijo continuation with same-mesh PMG	typically $10^{-2}/15$ unless a diagnostic states otherwise	$P_4(L_5)$ endpoint; $P_4(L_6)$–$P_4(L_7)$ fixed work	wall time
3D Mohr–Coulomb validation	endpoint-surrogate agreement, not path-history validation	fixed-$\lambda_{\rm sr}$ comparator diagnostics with matched boundary data	not used for timing comparison	fixed-$\lambda_{\rm sr}$ observables; strain/profile entries diagnostic	secondary to agreement metrics
3D Mohr–Coulomb performance	second-order routes, globalization, parallel scaling	Armijo, residual-bisection, or trust-region Newton with FGMRES/PMG	10^{-1} or 10^{-2} with max 100, as stated by configuration	relative correction, gradient target, cap, or fixed work	wall, solve, or solver-total time
Topology	distributed design-mechanics timing and rank consistency	adaptive reduced-objective continuation with PETSc mechanics and GAMG	mechanics FGMRES/GAMG: $10^{-4}/100$	design/state-change stall criterion or fixed schedule	end-to-end wall time

7.1 Globalization Method Comparison

Table 16 compares how nonlinear globalization policies behave on the tested fixed cases. Each comparison holds the element assembly path and preconditioner family fixed within the problem and varies only the nonlinear policy. Timed-out configurations are retained as capped solver behavior rather than convergence evidence.

The entries are representative fixed cases rather than one physical problem: p-Laplace uses a level-10 diagnostic case with one nonlinear solve, Ginzburg–Landau uses a level-10 fixed-budget comparison, hyperelasticity uses the first eight level-4 rank-local load steps, and 3D plasticity uses the $P_2(L_1)$ endpoint surrogate at $\lambda_{\rm sr} = 1.55$. The four configurations use 32, 16, 32, and 32 MPI ranks with 30, 120, 270, and 180 s caps.

The scalar and mechanics cases show different preferences. Newton with Armijo line search has the lowest reported time on the smooth p-Laplace diagnostic case, but times out on Ginzburg–Landau; the hyperelasticity excerpt completes under all three policies, with the hybrid reducing nonlinear iterations and reported solve time relative to line-search-only Newton.

The 3D plasticity endpoint is the most discriminating case in this suite for nonlinear-policy choice. Here and below, g denotes the free-DOF gradient of the merit functional used by the stated nonlinear solver. At the stricter $\|g\|_2 < 10^{-4}$ criterion, line-search-only Newton reaches the iteration cap while both trust-region policies converge, and the hybrid is the lower-time trust-region configuration. We therefore keep the Ginzburg–Landau level-10 fixed-budget check separate in Table 17: trust-region variants complete with matching energy and Newton/Krylov work, while the line-search-only configuration reports elapsed time and its cap.

7.2 Derivative-Route Comparison

Table 18 asks whether the second-order route changes the endpoint state or mainly changes the assembly and linearization cost. The three matched cases compare element AD, colored sparse recovery, and, where available, constitutive AD under the same nonlinear problem definition. The timing-scope column states whether the reported time is wall time, solve time, or a solver-internal total for the specific configuration.

Table 16: Fixed-case comparison of nonlinear globalization policies. LS denotes Armijo line search and TR denotes Steihaug trust region. The gradient-to-target column is reported for the 3D plasticity endpoint gate.

Outcome and endpoint value

Benchmark	Method	Ranks	Outcome	Completed/requested steps	Reported time [s]	Grad/target	Energy
p-Laplace L_{10}	Newton + LS	32	completed	1/1	6.95	–	−0.2517224496
p-Laplace L_{10}	Steihaug TR	32	completed	1/1	11.7	–	−0.2517227312
p-Laplace L_{10}	Hybrid TR+LS	32	completed	1/1	12	–	−0.2517227312
Ginzburg–Landau L_{10}	Newton + LS	16	timeout	0/1	121	–	–
Ginzburg–Landau L_{10}	Steihaug TR	16	completed	1/1	14.3	–	0.3456246783
Ginzburg–Landau L_{10}	Hybrid TR+LS	16	completed	1/1	14.3	–	0.3456246783
Hyperelasticity L_4	Newton + LS	32	completed	8/8	194	–	9.733145
Hyperelasticity L_4	Steihaug TR	32	completed	8/8	178	–	9.73101
Hyperelasticity L_4	Hybrid TR+LS	32	completed	8/8	149	–	9.730978
3D plasticity $P_2(L_1)$	Newton + LS	32	iteration cap	0/1	70.2	1.18	−3,012,813
3D plasticity $P_2(L_1)$	Steihaug TR	32	completed	1/1	20.8	0.00967	−3,012,813
3D plasticity $P_2(L_1)$	Hybrid TR+LS	32	completed	1/1	16.4	0.579	−3,012,813

Nonlinear and Krylov work

Benchmark	Method	Ranks	Newton	Krylov	LS evals	TR rejects
p-Laplace L_{10}	Newton + LS	32	15	32	15	0
p-Laplace L_{10}	Steihaug TR	32	29	32	0	0
p-Laplace L_{10}	Hybrid TR+LS	32	29	32	29	0
Ginzburg–Landau L_{10}	Newton + LS	16	0	0	0	0
Ginzburg–Landau L_{10}	Steihaug TR	16	19	25	0	0
Ginzburg–Landau L_{10}	Hybrid TR+LS	16	19	25	19	0
Hyperelasticity L_4	Newton + LS	32	420	6984	2758	0
Hyperelasticity L_4	Steihaug TR	32	314	7245	0	95
Hyperelasticity L_4	Hybrid TR+LS	32	295	6849	499	3
3D plasticity $P_2(L_1)$	Newton + LS	32	80	858	224	0
3D plasticity $P_2(L_1)$	Steihaug TR	32	13	96	0	1
3D plasticity $P_2(L_1)$	Hybrid TR+LS	32	11	73	12	0

Table 17: Ginzburg–Landau level-10 fixed-budget globalization comparison on eight MPI ranks. The configurations use the same element assembly and Hypre preconditioner; only the nonlinear globalization policy changes. The timed-out configuration reports elapsed wall time and the stated wall-time cap.

Method	Outcome	Newton	Krylov	LS evals	TR rejects	Reported time [s]	Cap [s]	Energy
Newton + LS	timeout	–	–	–	–	301	300	–
Steihaug TR	completed	19	27	0	0	28.5	–	0.3456246783
Hybrid TR+LS	completed	19	27	19	0	28.5	–	0.3456246783

Across all three benchmark cases, colored sparse recovery reaches the same table-reported energy, up to displayed precision, and identical nonlinear/Krylov iteration counts as element AD. The scalar and hyperelasticity cases show a wall-time premium for colored sparse recovery. On the 3D plasticity endpoint, constitutive AD has the lowest reported time, while element AD and colored sparse recovery have matching iteration counts and comparable solve times. The colored sparse-recovery route requires 240–420 colors across the 32 ranks; its comparable solve time is consistent with batched rank-local HVPs having nearly the same measured Hessian-evaluation cost as element AD on this $P_2(L_1)$ case.

Table 19 reports one-Newton Hessian-evaluation cost. At the same 79,024 free DOFs, moving from $P_1(L_2)$ to $P_2(L_1)$ raises colors from 66–90 to 240–420, Hessian time from 0.332 to 1.412 s, and rank-maximum resident set size (RSS) from about 0.6 to 1.1 GiB; the P_4 colored-recovery case is outside this table.

Table 18: Matched-case derivative-route comparison. Colored sparse recovery uses rank-local AD-HVP graph-color groups; color counts are shown where available.

Outcome and timing

Benchmark	Route	Outcome	Time [s]	Timing scope	Energy
p-Laplace L_9	Element AD	completed	1.14	solver-internal	-7.960003
p-Laplace L_9	Colored recovery (AD-HVP)	completed	1.45	solver-internal	-7.960003
Hyperelasticity L_4 step 1	Element AD	completed	11.7	solve	0.1519923275
Hyperelasticity L_4 step 1	Colored recovery (AD-HVP)	completed	34.6	solve	0.1519923275
3D plasticity $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Element AD	completed	16.3	solve	$-3{,}012{,}813$
3D plasticity $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Colored recovery (AD-HVP)	completed	16.4	solve	$-3{,}012{,}813$
3D plasticity $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Constitutive AD	completed	4.81	solve	$-3{,}012{,}813$

Work and Hessian construction

Benchmark	Route	Ranks	Newton	Krylov	Hessian [s]	Color groups
p-Laplace L_9	Element AD	32	6	11	–	–
p-Laplace L_9	Colored recovery (AD-HVP)	32	6	11	–	–
Hyperelasticity L_4 step 1	Element AD	32	22	492	–	–
Hyperelasticity L_4 step 1	Colored recovery (AD-HVP)	32	22	492	–	–
3D plasticity $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Element AD	32	11	73	6.57	–
3D plasticity $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Colored recovery (AD-HVP)	32	11	73	6.59	240–420
3D plasticity $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Constitutive AD	32	11	73	1.6	–

Table 19: 3D plasticity one-Newton derivative-route cost versus discretization. Elem DOFs is the Hessian-evaluation vector size; overlap DOFs is the maximum local overlap vector size during assembly.

Discretization size

Element	Route	Work	Free DOFs	Elem. DOFs	Overlap DOFs	RSS [GiB]
$P_1(L_1)$	Element AD	one Newton step	10,526	12	675	0.5
$P_1(L_1)$	Colored recovery (AD-HVP)	one Newton step	10,526	12	675	0.5
$P_1(L_1)$	Constitutive AD	one Newton step	10,526	12	675	0.5
$P_1(L_2)$	Element AD	one Newton step	79,024	12	3624	0.6
$P_1(L_2)$	Colored recovery (AD-HVP)	one Newton step	79,024	12	3624	0.6
$P_1(L_2)$	Constitutive AD	one Newton step	79,024	12	3624	0.5
$P_2(L_1)$	Element AD	one Newton step	79,024	30	3933	0.9
$P_2(L_1)$	Colored recovery (AD-HVP)	one Newton step	79,024	30	3933	1.1
$P_2(L_1)$	Constitutive AD	one Newton step	79,024	30	3933	0.6
$P_4(L_1)$	Element AD	one Newton step	610,964	105	27177	1.2
$P_4(L_1)$	Constitutive AD	one Newton step	610,964	105	27177	1.1

Linearization cost

Element	Route	Free DOFs	Krylov	Solve [s]	Hessian [s]	Colors
$P_1(L_1)$	Element AD	10,526	11	0.0806	0.257	–
$P_1(L_1)$	Colored recovery (AD-HVP)	10,526	11	0.0704	0.256	57–90
$P_1(L_1)$	Constitutive AD	10,526	11	0.0706	0.217	–
$P_1(L_2)$	Element AD	79,024	3	0.232	0.322	–
$P_1(L_2)$	Colored recovery (AD-HVP)	79,024	3	0.273	0.332	66–90
$P_1(L_2)$	Constitutive AD	79,024	3	0.164	0.257	–
$P_2(L_1)$	Element AD	79,024	5	0.83	1.31	–
$P_2(L_1)$	Colored recovery (AD-HVP)	79,024	5	0.827	1.41	240–420
$P_2(L_1)$	Constitutive AD	79,024	5	0.304	0.535	–
$P_4(L_1)$	Element AD	610,964	13	8.48	7.04	–
$P_4(L_1)$	Constitutive AD	610,964	13	3.97	3.62	–

7.3 p-Laplace

The p-Laplace problem definition is given in Section 5.1, with discrete problem (8). In this scalar setting, the derivative-route evidence is direct: exact element Hessians remain effective in the reported distributed configurations, while colored sparse recovery reaches nearly identical discrete energies at higher wall time.

In the finest plotted case, the JAX+PETSc element-AD path reports 1.14 s on 32 ranks, while colored sparse recovery reports 1.45 s with the same Newton and Krylov iteration counts. For the matched 32-rank case summarized in Table 18, the displayed energies match, so this block mainly measures the overhead of sparse recovery on a smooth scalar problem. Across the plotted ranks, element AD drops from 10.13 s on one rank to 1.14 s on 32 ranks; colored sparse recovery follows the same trend with persistent recovery overhead. The FEniCS curve in the same figure provides

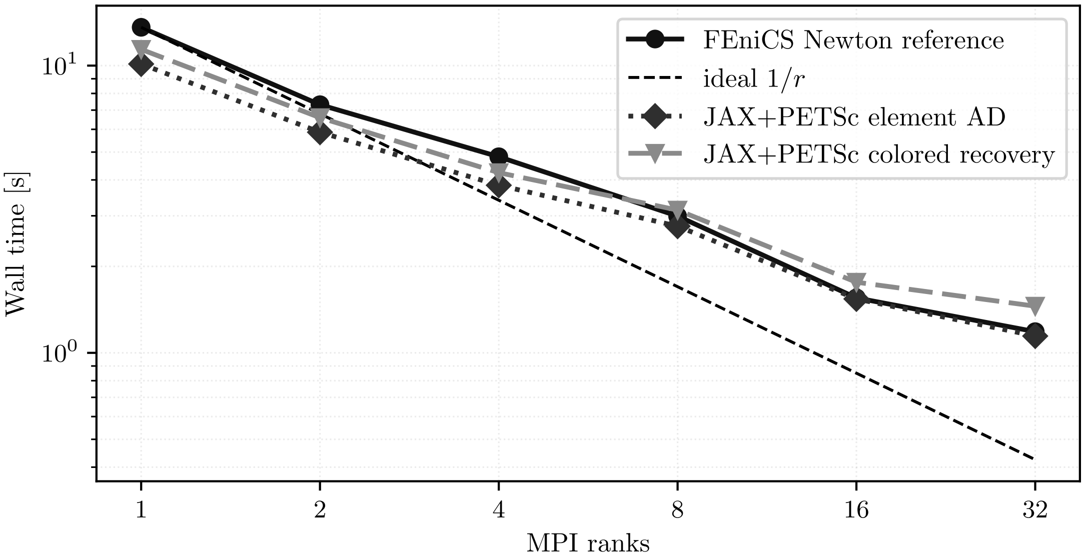


Figure 21: p-Laplace strong scaling on the finest tested grid. All curves correspond to the same discrete problem from (8).

reference-implementation context; the comparison above isolates the two JAX+PETSc derivative routes and is not interpreted as a backend performance ranking.

7.4 Ginzburg–Landau

The Ginzburg–Landau family from Section 5.2, with discrete problem (12), shows the same high-level derivative story under a more indefinite linearization. The computational point is not a dramatic separation between element-Hessian and colored sparse-recovery routes, but that the same sparse infrastructure handles the reported nonconvex scalar-energy configurations.

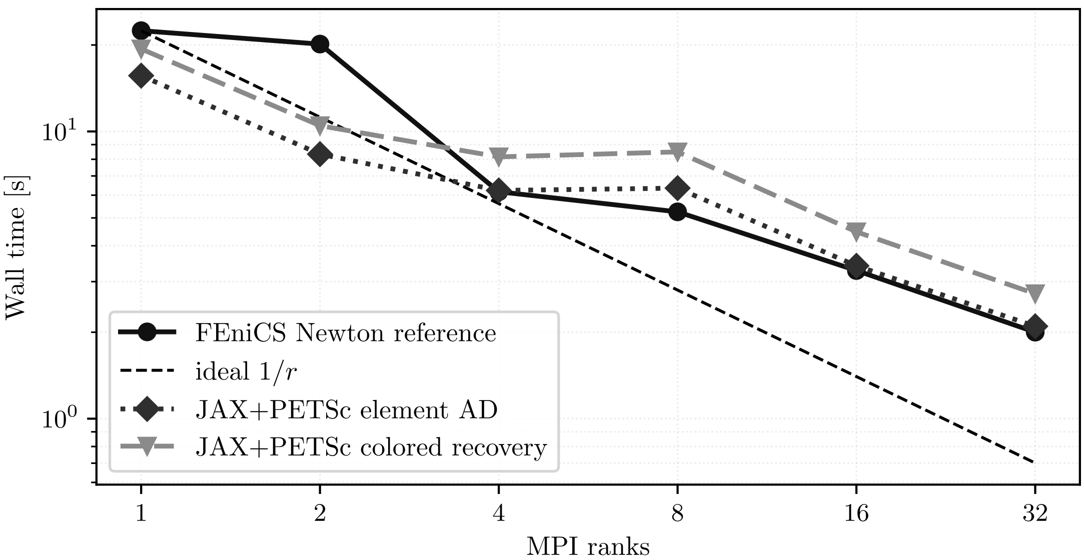


Figure 22: Ginzburg–Landau strong scaling for the tested fine-grid case governed by (12).

At 32 ranks, the element-AD and colored sparse-recovery JAX+PETSc configurations report 2.10 s and 2.72 s, respectively. Figure 9 and table 8 give the representative energy-agreement evidence for this family, while Figure 22 reports timing only. Thus the main visible difference in the scaling plot is the colored sparse-recovery wall-time premium on this nonconvex scalar energy. Across the plotted ranks, element AD falls from 15.64 s on one rank to 2.10 s on 32 ranks, with a nonideal plateau around 4–8 ranks before the 32-rank point. The FEniCS curve is included as reference-implementation context, while the paragraph above isolates the JAX+PETSc derivative-route cost rather than a backend performance ranking.

7.5 Hyperelasticity

Section 5.3 defines the boundary data and (16). Here the full load path tests globalization and preconditioning; derivative-route evidence is limited to the first-step replicated-element case. The distributed implementation completes the load path, Figure 23 reports completed-path timing, and the JAX-FEM comparison remains in Section 6.2.

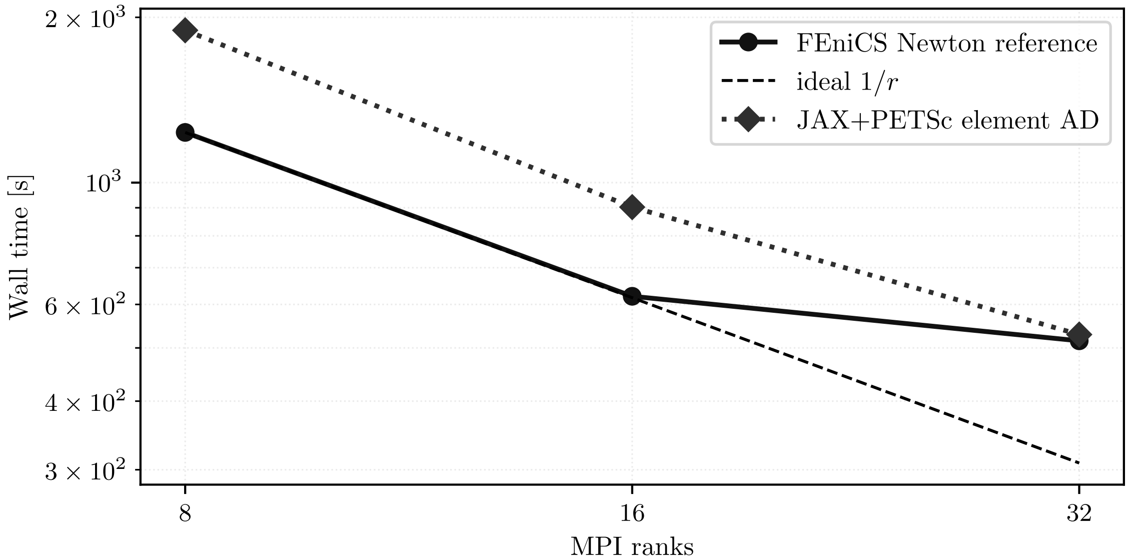


Figure 23: Hyperelasticity completed-path timing for the 24-step load path defined by (16).

The 24-step hyperelasticity completed-path timing curve reduces JAX+PETSc element-AD wall time from 1894 s at 8 ranks to 529 s at 32 ranks, a factor of 3.58 over a fourfold rank increase. This motivates the more targeted fixed-work distribution and PMG diagnostics below.

The fixed-work multi-node experiment uses the same trust-region Newton path as Section 5.3 and asks how rank-local level-5 first-step assembly behaves as ranks and PMG coarse-solver layouts vary. Runs use one MPI rank per core on dual-socket AMD EPYC 7H12 nodes [39], local CPU and memory binding, rank-local elements, coordinate-format preallocation, point-to-point overlap exchange, Chebyshev/Jacobi-smoothed PMG with coarsest level L_2, and redundant-LU/MUMPS coarse solves. Figure 24 and table 22 report solver-internal time; all listed configurations have 21 Newton and 43 Krylov iterations.

Table 20 reports the memory evidence that supports using the rank-local assembly variant in the larger CPU sweeps. The fixed-work agreement entries compare the replicated and rank-local level-4 first-step solve at four MPI ranks; the one-linearization memory entries use one Newton linearization on the level-5 mesh at 8, 16, and 32 ranks. The table reports energy, operating-system peak resident set size (RSS), and the measured footprint of assembly arrays, making clear whether a configuration stores global mesh and adjacency data or only local elements plus overlap.

Table 20: Rank-local hyperelasticity memory and overlap evidence. RSS max/sum gives the maximum and summed per-rank resident set size. Tracked sum is the reported assembly-array footprint, and overlap/owned is the summed rank-local overlap DOF count divided by global owned free DOFs for the rank-local overlap assembly.

Solve work

Purpose	Assembly layout	Outcome	Level	Ranks	Newton	Krylov	Solve [s]	Energy
fixed-work agreement	replicated	completed	L_4	4	26	571	51.2	0.151971
fixed-work agreement	rank-local	completed	L_4	4	26	571	50.8	0.151971
one-linearization memory	replicated	single linearization	L_5	8	1	1	10.7	53.95804
one-linearization memory	rank-local	single linearization	L_5	8	1	1	10.5	53.95926
one-linearization memory	replicated	single linearization	L_5	16	1	1	5.57	32.927343
one-linearization memory	rank-local	single linearization	L_5	16	1	1	5.5	32.927867
one-linearization memory	replicated	single linearization	L_5	32	1	1	3.37	23.942407
one-linearization memory	rank-local	single linearization	L_5	32	1	1	3.36	23.803027

Memory and overlap

Purpose	Assembly layout	Level	Ranks	RSS max/sum [GiB]	Tracked sum [GiB]	Overlap/owned
fixed-work agreement	replicated	L_4	4	3.6/14.2	5.49	1.01
fixed-work agreement	rank-local	L_4	4	3.3/13.2	6.01	1.01
one-linearization memory	replicated	L_5	8	14.9/118.7	44.34	1.01
one-linearization memory	rank-local	L_5	8	12.3/98.3	47.87	1.01
one-linearization memory	replicated	L_5	16	9.4/150.9	45.45	1.03
one-linearization memory	rank-local	L_5	16	6.3/100.3	47.89	1.03
one-linearization memory	replicated	L_5	32	6.7/215.6	47.65	1.05
one-linearization memory	rank-local	L_5	32	3.3/104.9	47.92	1.05

The level-4 fixed-work entries show matching Newton and Krylov counts and the same rounded energy value for replicated and rank-local assembly. On the larger level-5 one-linearization entries, rank-local assembly reduces summed

RSS from 215.6 GiB to 104.9 GiB at 32 ranks, with solve times 3.37 s and 3.36 s. The level-5 energy values are fixed-work endpoint states after one linearization, so these entries are memory/cost diagnostics rather than endpoint-agreement checks.

Table 21 isolates the first-step level-5 PMG choice on the same 32-rank level-5 problem with fixed nonlinear work. All configurations perform the same eight Newton linearizations, so the table compares the cost and endpoint state reached under different preconditioner and coarse-solver policies. The energy column is the state after this fixed nonlinear work, so Table 21 is a solver-policy comparison rather than an equal-accuracy timing comparison or endpoint-accuracy ranking.

Table 21: Rank-local hyperelasticity level-5 PMG and coarse-solver sensitivity at 32 MPI ranks with fixed nonlinear work. The PMG configurations use the same rank-local element path as the scaling configurations.

Outcome and solver work

Precond.	Nonlinear work	Newton	Krylov	Solver [s]	Energy
GAMG	8 Newton linearizations	8	42	22.3	1.608441
PMG L_2 + Hypre	8 Newton linearizations	8	92	23.3	0.5252349427
PMG L_2 + MUMPS	8 Newton linearizations	8	17	15.3	0.3848912708
PMG L_3 + MUMPS	8 Newton linearizations	8	17	16.3	0.3864166034

Time breakdown and coarse solve

Precond.	Coarse	Assembly [s]	PC [s]	Linear [s]
GAMG	–	9.11	6.41	3.91
PMG L_2 + Hypre	Hypre	10.3	2.93	8.14
PMG L_2 + MUMPS	MUMPS, one redundant group	9.13	2.55	1.91
PMG L_3 + MUMPS	MUMPS, one redundant group	9.14	2.93	2.51

The fixed-work table reports that the MUMPS-coarse PMG variants reduce total Krylov iterations from 92 with Hypre coarse solves to 17, and reduce solver time from 23.3 s to about 16 s. This is preconditioner-policy evidence for a fixed nonlinear workload, not a statement about final load-path accuracy. The energy differences in the table are fixed-work endpoint-state differences after the same eight Newton linearizations.

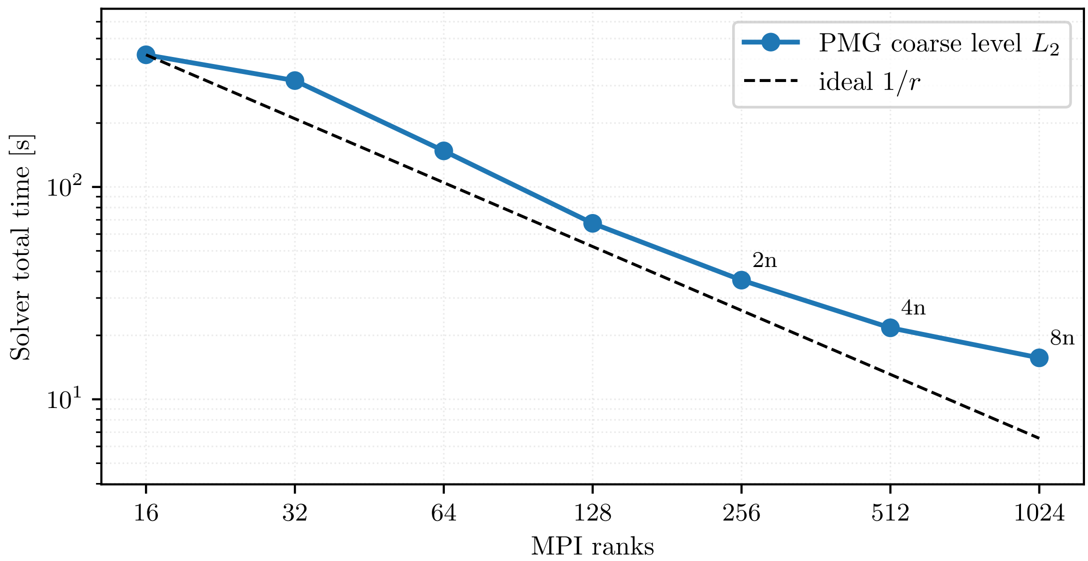


Figure 24: Single- and multi-node CPU level-5 hyperelasticity first-step scaling at fixed nonlinear work with the PMG coarse level fixed at L_2. Annotations show CPU-node counts.

Table 22: Single- and multi-node CPU level-5 hyperelasticity first-step timings used in figure 24. This is a fixed-work first-step timing study with a common PMG hierarchy, not a full load-path comparison.

Ranks	Nodes	Coarse groups	Ranks/group	Solver total [s]	First step [s]	Newton iters	Krylov iters
16	1	1	16	419	367	21	43
32	1	1	32	317	280	21	43
64	1	1	64	148	132	21	43
128	1	1	128	67.4	58.4	21	43
256	2	2	128	36.4	30.7	21	43
512	4	4	128	21.7	17.7	21	43
1024	8	8	128	15.7	12.3	21	43

Figure 24 and table 22 show a fixed-work timing drop from 419 s on 16 ranks to 15.7 s on 1024 ranks, with the same 21 Newton iterations and 43 Krylov iterations in all listed configurations. This evidence isolates the multi-node sparse-solver and PMG-coarse-level behavior for one load step, rather than the accuracy of the full load path.

7.6 2D Mohr–Coulomb Plasticity

The 2D Mohr–Coulomb plasticity model is defined in Section 5.4, with Davis-B reduction (19) and discrete surrogate (20). For this family, the most informative computational evidence in the present paper is not a single strong-scaling curve. The benchmark section separates the completed endpoint case from the larger fixed-cost studies in Figure 13 and table 10, and the reference-continuation comparison is collected later in Section A.

The $P_4(L_5)$ case is the endpoint evidence for this family. The completed endpoint has energy -212.538 and wall time 147 s. The larger $P_4(L_6)$ and $P_4(L_7)$ cases are fixed-work diagnostics for nonlinear-tail and PMG behavior; both report energy -212.539 after 20-iteration caps, with wall times 147 s and 253 s.

The 8-rank and 32-rank reference-continuation entries in Table 31 support solver-policy timing under a common continuation logic. They are not a separate validation result or a strong-scaling claim. In this diagnostic, increasing from 8 to 32 ranks reduces wall time from 489 s to 198 s, while the continuation Krylov totals differ, 733 versus 829.

7.7 3D Mohr–Coulomb Plasticity

The governing problem is defined in Section 5.5, with Davis-B reduction (24), scalar constitutive potential (32), discrete surrogate (33), and deviatoric-strain norm (34). The validation limits for this endpoint surrogate are reported in Section 6.3. With that interpretation fixed, this section asks two performance questions: which second-order route has the lowest reported cost on the fixed high-order derivative-route case under matched endpoint observables, and how the endpoint surrogate behaves across degree, resolution, and rank count.

Derivative-route comparison. The first question is whether the second-order route changes the endpoint observables or only the cost of derivative construction. On the fixed $P_4(L_1)$ case at $\lambda_{\mathrm{sr}} = 1.5$ and 8 MPI ranks, Figure 25 shows timing and linear-iteration costs, while Table 23 reports the same Newton counts, total Krylov counts, and endpoint observables to displayed precision across the three routes. In the table, ω denotes the external-work observable and $u_{\max}$ denotes the maximum displacement magnitude.

The observed differences are timing differences under this fixed solver policy. Constitutive AD has the lowest reported wall time on this fixed case, with wall time 222 s and solve time 191 s, compared with 288 s/255 s for element AD and 372 s/253 s for colored sparse recovery. This residual-bisection Newton comparison uses the Hypre-backed rank-local linear-solver profile; it is therefore separate from the MUMPS-backed PMG configurations used by the later $\lambda_{\mathrm{sr}} = 1.55$ convergence studies.

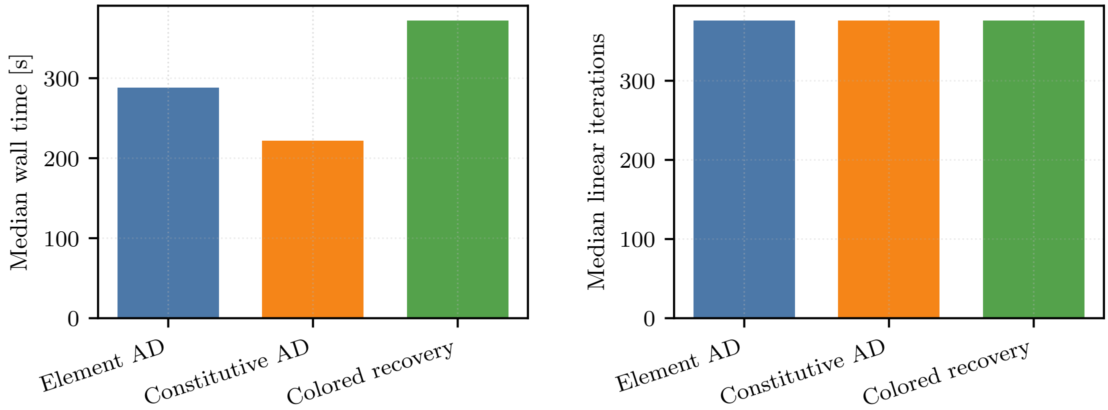


Figure 25: Fixed 3D plasticity derivative-route comparison on $P_4(L_1)$, $\lambda_{sr} = 1.5$, and 8 MPI ranks.

Table 23: 3D plasticity derivative-route comparison on the fixed $P_4(L_1)$, $\lambda_{sr} = 1.5$, 8-rank case. These configurations use the Hypre-backed rank-local profile and are separate from the MUMPS-backed PMG configurations in the $\lambda_{sr} = 1.55$ studies.

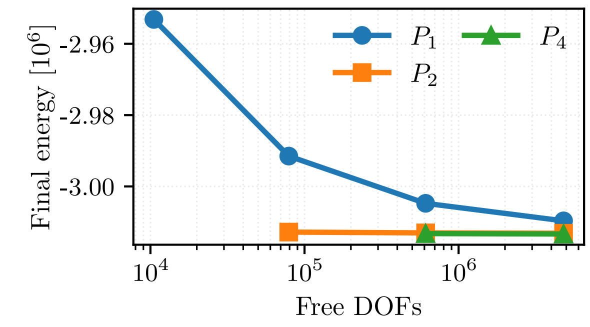
Timing and work

Route	Wall time [s]	Solve time [s]	Newton iters	Krylov iters
Element AD	288	255	8	376
Constitutive AD	222	191	8	376
Colored recovery (AD-HVP)	372	253	8	376

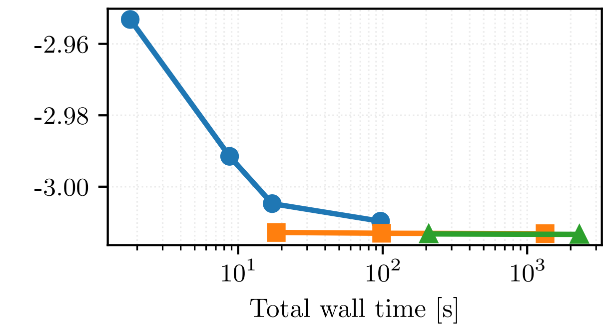
Endpoint observables

Route	Energy	ω	$u_{\max}$
Element AD	−3,010,861	6,027,722	0.731848
Constitutive AD	−3,010,861	6,027,722	0.731848
Colored recovery (AD-HVP)	−3,010,861	6,027,722	0.731848

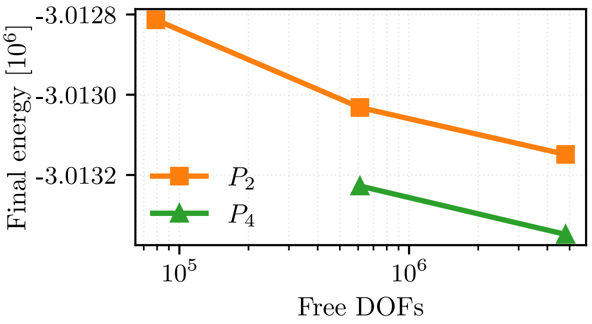
Discretization and scaling. The remaining figures separate endpoint-surrogate behavior across degree/resolution choices from timing-only scaling on the selected $P_4(L_2)$ configuration. The zoomed panels in Figure 26 expose the near-overlap hidden by coarse P_1 variation. At matched DOF counts, the solver reaches similar endpoint energies along both p- and h-refined paths; higher order on coarser meshes lowers the displayed energy in this nine-case endpoint study at a clear wall-time premium.



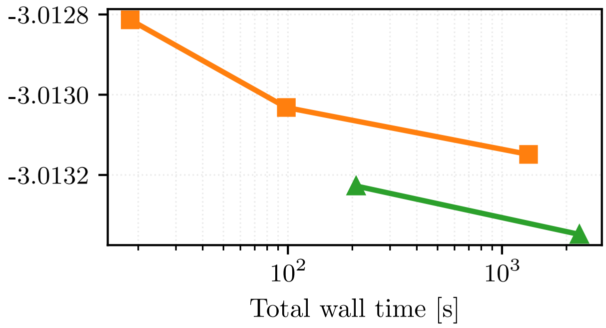
(a) All degrees: final energy versus free DOFs.



(b) All degrees: final energy versus total wall time.



(c) P_2/P_4 zoom: final energy versus free DOFs.



(d) P_2/P_4 zoom: final energy versus total wall time.

Figure 26: Bottom-clamped 3D plasticity degree-versus-resolution study at $\lambda_{\text{sr}} = 1.55$. The top panels show the full $P_1/P_2/P_4$ comparison, while the bottom panels zoom to P_2/P_4 so the near-overlap hidden by the coarse P_1 points becomes visible. The left column plots final values of (33) against free DOFs; the right column plots the same converged energies against end-to-end wall time. All timings use the same 32-rank configuration.

Table 24: Strong-scaling timing study for the 3D plasticity $P_4(L_2)$ case at $\lambda_{\text{sr}} = 1.0$ using the constitutive-AD PMG configuration. These timings provide solver evidence for a separate strength-reduction factor from the bottom-clamped $\lambda_{\text{sr}} = 1.55$ benchmark figures. The stopping target is $\|g\| < 0.01$, the KSP relative tolerance is 0.1, and the Newton cap is 50 iterations.

Ranks	Wall time [s]	Speedup	Newton iters	Krylov iters	Stop target	Status
1	5424	1	17	114	$\|g\| < 0.01$	completed
2	2914	1.861	17	114	$\|g\| < 0.01$	completed
4	1717	3.16	17	114	$\|g\| < 0.01$	completed
8	1164	4.658	17	114	$\|g\| < 0.01$	completed
16	608	8.916	17	114	$\|g\| < 0.01$	completed
32	300	18.053	17	114	$\|g\| < 0.01$	completed

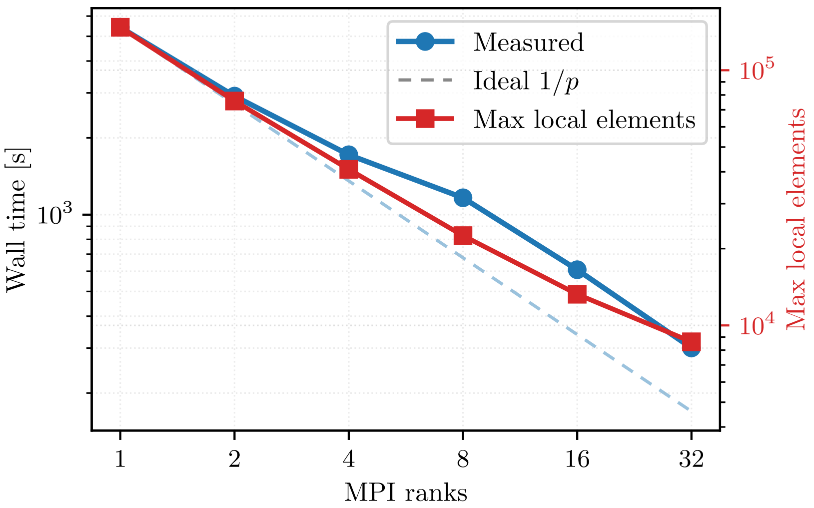


Figure 27: Strong-scaling timing study for the 3D plasticity $P_4(L_2)$ case at $\lambda_{\mathrm{sr}} = 1.0$ using the constitutive-AD PMG configuration. This is a separate timing study from the bottom-clamped $\lambda_{\mathrm{sr}} = 1.55$ results shown in Figures 14, 15 and 26.

Figure 27 and table 24 give a separate $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.0$ timing result for the Hypre-coarse PMG variant: matched iteration counts on 1–32 ranks and end-to-end time dropping from 5424 s to 300 s. Additional reference-formula assembly and continuation evidence is reported later in Section A.

The converged bottom-clamped $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.55$ comparison instead uses constitutive-AD rank-local assembly with MUMPS-backed $P_4 \to P_2 \to P_1$ PMG and the same observed nonlinear/Krylov work; all configurations are marked completed and report 29 Newton iterations, 1240 Krylov iterations, and the same displayed energy. With $\|g\|_{\mathrm{stop}}/\|g\|_{\mathrm{target}}$ denoting the final stopping gradient divided by its one-percent-of-initial-gradient target, Figure 28 and table 25 report CPU timings together with the gradient-target ratio used for the stopping-gradient check.

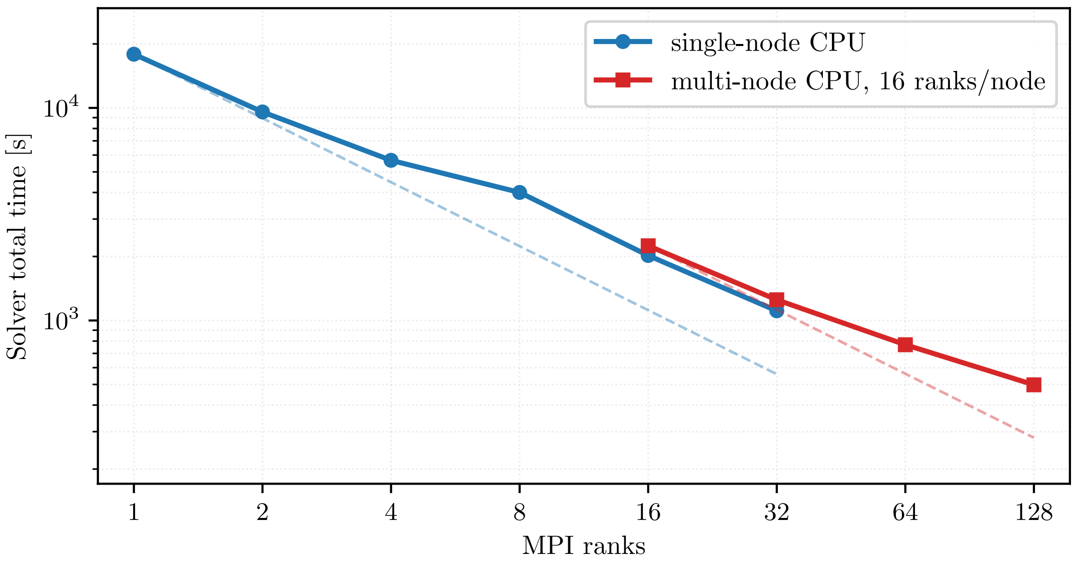


Figure 28: Single-node and multi-node CPU converged scaling for the bottom-clamped 3D plasticity $P_4(L_2)$ case at $\lambda_{\mathrm{sr}} = 1.55$. Dashed references are ideal $1/r$ lines anchored separately to the first point of each CPU setting.

Table 25: Converged timings for the 3D plasticity $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.55$ configuration shown in figure 28. Single-node configurations use one CPU node; multi-node configurations use 16 ranks per CPU node. The nonlinear stop requires relative correction below 0.002 and final gradient below one percent of the first Newton gradient. The stopping-gradient norm in this table reports the relative-gradient stopping ratio for this scaling comparison, not the absolute final-gradient column in the degree benchmark table.

CPU setting	Ranks	Nodes	Solver total [s]	Newton iters	Krylov iters	Energy	$\|g\|_{\mathrm{stop}}$/target	Status
single-node CPU	1	1	17,890	29	1240	−3,013,348	0.525	completed
single-node CPU	2	1	9574	29	1240	−3,013,348	0.525	completed
single-node CPU	4	1	5667	29	1240	−3,013,348	0.525	completed
single-node CPU	8	1	4003	29	1240	−3,013,348	0.525	completed
single-node CPU	16	1	2022	29	1240	−3,013,348	0.525	completed
single-node CPU	32	1	1110	29	1240	−3,013,348	0.525	completed
multi-node CPU	16	1	2246	29	1240	−3,013,348	0.525	completed
multi-node CPU	32	2	1251	29	1240	−3,013,348	0.525	completed
multi-node CPU	64	4	769	29	1240	−3,013,348	0.525	completed
multi-node CPU	128	8	498	29	1240	−3,013,348	0.525	completed

The distributed layout measurements in Table 26 identify one observed limit on ideal scaling. The maximum local overlap vector shrinks with rank count, but the sum of rank-local overlap DOFs increases from about the owned problem size on one rank to more than four times the owned size on 128 ranks. This overlap replication comes from subdomain boundaries and high-order element support, so it is a decomposition cost rather than duplicated global mesh storage.

Table 26: Distributed layout measurements for the converged 3D plasticity scaling comparison. "Max local DOFs" is the largest local-overlap vector size on any rank; "Overlap / owned DOFs" is the summed rank-local overlap DOF count divided by the global owned free-DOF count.

CPU setting	Ranks	Nodes	Max local DOFs	Overlap / owned DOFs
single-node CPU	1	1	4,859,595	1.01
single-node CPU	2	1	2,495,265	1.05
single-node CPU	4	1	1,284,351	1.11
single-node CPU	8	1	669,783	1.21
single-node CPU	16	1	351,219	1.41
single-node CPU	32	1	182,613	1.83
multi-node CPU	16	1	351,219	1.41
multi-node CPU	32	2	182,613	1.83
multi-node CPU	64	4	91,821	2.66
multi-node CPU	128	8	50,637	4.31

7.8 Topology Optimization

The frozen reduced design problem and objective history are defined in Section 5.6, with mechanics subproblem (36) and frozen reduced objective (37). The numerical question here is distributed outer-loop stability: the mechanics solve and the design update must remain synchronized under continuation in p_{SIMP} and move-regularized updates. Because the outer-loop work can vary with the continuation schedule, we report end-to-end timing under adaptive stopping rather than strong scaling at a fixed computational workload.

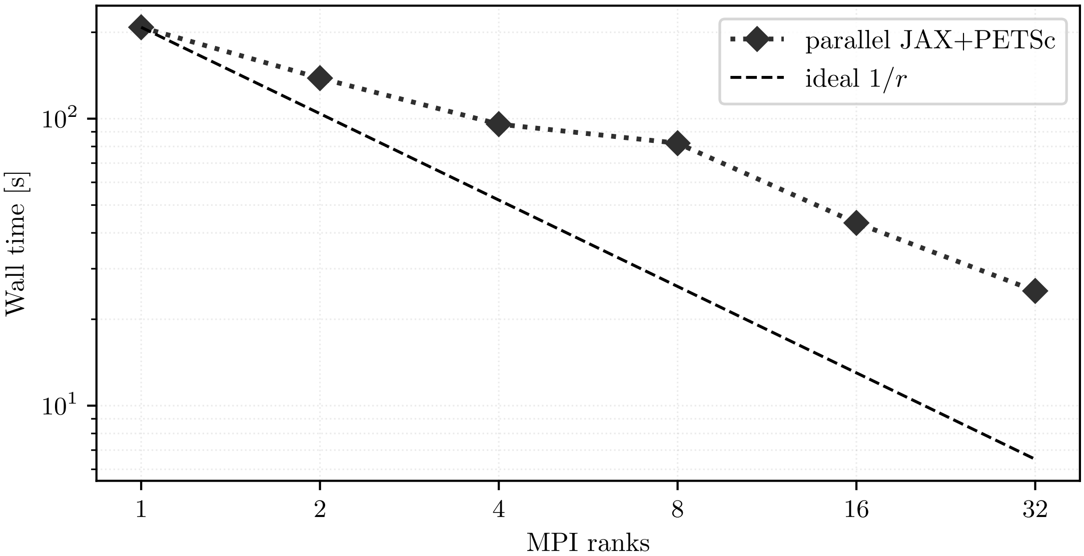


Figure 29: Topology end-to-end timing for the fine-grid reduced-design problem (37); Table 27 reports the per-rank stopping behavior.

Table 27: Parallel topology endpoint observables and timing for the fine-grid reduced-design study. Wall time is adaptive end-to-end timing, not fixed-work strong scaling; the table reports the adaptive schedule and completion status explicitly.

Ranks	Wall time [s]	Outer iters	p_{SIMP}	Compliance	Volume	Schedule	Status
1	208	65	4.2	9.1559	0.3884	adaptive	completed
2	138	72	5.6	8.9473	0.3932	adaptive	completed
4	95.7	69	4.6	9.1684	0.385	adaptive	completed
8	82.1	67	7	9.2178	0.3932	adaptive	completed
16	43.3	66	5.8	9.6859	0.3798	adaptive	completed
32	25.1	66	6.6	9.9074	0.3749	adaptive	completed

Figure 29 and table 27 report fine-grid wall time decreasing from 208 s on one rank to 25.1 s on 32 ranks. Outer iterations, final continuation exponents, compliance (8.9473–9.9074), and achieved volume fractions (0.3749–0.3932) vary across ranks. Thus this is adaptive end-to-end behavior under the multiplier and stopping rule, not target-volume equality, fixed-work strong scaling, or a rank-invariant endpoint.

Table 28 adds a controlled 40-iteration check on a smaller 384×192 grid: fixed initial design, disabled early/stall stopping, and smoother continuation ($p_{\mathrm{SIMP}} \leftarrow p_{\mathrm{SIMP}} + 0.1$). Let $\Theta_r = (\theta_h^{(r)}(x_i))_{i \in \mathcal{D}_h}$ be the active design-node density vector on the common grid for the r-rank run. Relative to the one-rank reference, all ranks complete 40 iterations, with compliance difference below 0.2% and density relative grid ℓ_2 difference $\|\Theta_r - \Theta_1\|_2 / \|\Theta_1\|_2$ below 0.04; these are observed discrepancies for this controlled case, not general rank-invariance thresholds.

Table 28: Topology controlled 40-iteration rank-consistency check. Each rank count completes the same 40 outer iterations; compliance and density differences are relative to the one-rank case.

Ranks	Schedule	Outer	Solve [s]	Compliance	Volume	p_{SIMP}	Rel. compliance diff.	Density rel. grid ℓ_2
1	fixed schedule	40	48.3	15.56	0.3085	5	0	0
2	fixed schedule	40	28.5	15.56	0.3083	5	1.29×10^{-6}	1.37×10^{-2}
4	fixed schedule	40	19.7	15.5308	0.3086	5	1.87×10^{-3}	1.91×10^{-2}
8	fixed schedule	40	16	15.5665	0.3081	5	4.16×10^{-4}	3.68×10^{-2}
16	fixed schedule	40	8.23	15.5606	0.3084	5	4.15×10^{-5}	1.47×10^{-2}
32	fixed schedule	40	6.43	15.5637	0.3083	5	2.41×10^{-4}	9.71×10^{-3}

The topology case extends the evidence from state solves to coupled design–mechanics optimization. Here the primary distributed JAX+PETSc realization is less about exact Hessians than about stable distributed design-gradient exchange and mechanics coupling, while pure JAX remains the compact serial design demonstration discussed earlier in Table 5.

7.9 Synthesis of the Computational Evidence

The results above provide the evidence blocks used in Section 8. The globalization tests separate problem-dependent nonlinear policies; the derivative-route comparisons isolate cost under fixed endpoint observables; the distribution, PMG, and partitioning studies expose solver-policy and sparse-ownership effects; and the topology study closes with an end-to-end design-and-mechanics loop plus a controlled rank-consistency check. Together, these measurements show which coupled choices are effective in the tested cases and where solver policy remains visible: local differentiable constitutive or energy evaluations can be coupled to PETSc-owned sparse nonlinear and linear solves across the tested scalar PDE, mechanics, plasticity, and topology cases, with comparator evidence scoped by problem family.

8 Discussion

The numerical evidence supports a methodological conclusion for the tested benchmark classes: in nonlinear finite-element energy minimization, the useful unit of algorithmic design is not the AD route alone. Local derivative construction, sparse ownership, nonlinear globalization, Krylov algebra, and preconditioning together determine the observed solver behavior. We therefore use a staged evidence base: formulation checks and matched endpoint-state comparisons support validation claims, while fixed-work route studies and distributed timing evidence support performance and solver-policy claims.

Derivative information is realized through the nonlinear solve. The high-order three-dimensional plasticity derivative-route ablation is the fixed-work example with the most complete derivative-route separation; see Figure 25 and table 23. On this fixed endpoint problem, the three derivative routes preserve the reported endpoint observables and iteration counts. The comparison therefore isolates derivative-route cost only after the residual, Jacobian, preconditioner, and nonlinear stopping policy have been fixed.

In this setting, constitutive AD has the lowest measured wall time, while colored sparse recovery remains a useful comparison and fallback route on the tested fixed-size problems. The conclusion is therefore not that one derivative route is uniformly best, but that derivative placement should be assessed inside the assembled sparse nonlinear solve that will use it.

Agreement evidence depends on matched definitions. The hyperelastic JAX-FEM comparison in Figure 19 and table 13 is useful because the mesh, displacement schedule, and endpoint-energy post-comparison are explicitly matched. The three-dimensional plasticity comparison in Figure 20 and table 14 uses a different agreement criterion: the highest-successful-load value on the tested grid, the $u_{\max}(\lambda_{\mathrm{sr}})$ curve, and endpoint displacement satisfy the stated criteria, while deviatoric-strain and upper-slope profile comparisons serve as diagnostics. These two examples illustrate the same validation rule: external or surrogate comparisons are informative when the problem definition and compared observables are named before agreement is interpreted. The three-dimensional plasticity endpoint result is therefore endpoint-scoped mechanics evidence, not a path-consistent reproduction of the Sysala-family incremental-history plasticity references [2–5] or the Čermák–Sysala–Valdman MATLAB implementation context [6].

Scalability evidence reflects solver policy. The distribution and PMG studies show that sparse ownership, overlap size, coarse solves, and preconditioner choices enter the numerical method. The $\lambda_{\mathrm{sr}} = 1.0$ three-dimensional plasticity strong-scaling study is timing-only evidence, whereas the bottom-clamped $\lambda_{\mathrm{sr}} = 1.55$ scaling sweep is the converged benchmark comparison. The topology study reports end-to-end timing under adaptive stopping, then adds a smaller fixed-schedule rank-consistency check to separate distributed design-gradient exchange from adaptive termination effects. The frozen-preconditioner PMG diagnostics should similarly be read as solver-sensitivity tests, not as external baselines.

Together, these distinctions delimit the contribution. We present a nonlinear FEM toolset that couples JAX-based local differentiation with PETSc sparse distributed solvers, and we evaluate that coupling with comparison designs that specify their own limits. The results support the toolset as an evaluation framework for the tested benchmark classes: derivative route, assembly, globalization, Krylov algebra, and preconditioning are reported as coupled numerical choices. They do not imply that a single backend, globalization policy, preconditioner, or derivative construction dominates nonlinear finite-element computations in general.

9 Conclusions

We developed a JAX+PETSc toolset for nonlinear finite-element energy minimization that couples local differentiable energy models to PETSc sparse distributed nonlinear solvers. Across the tested problems, the numerical evidence supports treating derivative construction, sparse ownership, nonlinear globalization, Krylov algebra, and preconditioning as one solver design rather than as separable implementation details. The concrete contribution is therefore not a single winning backend or preconditioner, but a scientific toolset and evaluation protocol: fixed-model derivative-route comparisons, PETSc-owned sparse MPI assembly, scoped validation checks, and timing studies whose solver-policy assumptions are reported explicitly.

Key evidence comes from the high-order 3D plasticity derivative-route comparison, the endpoint-scoped hyperelasticity and plasticity validation checks, the distributed mechanics solver-policy studies, and the topology rank-consistency diagnostic. These evidence blocks support the JAX+PETSc toolset for the benchmark families tested here without implying a universal backend, preconditioner, globalization, or derivative-route ranking. Pure JAX and FEniCS checks, the JAX-FEM endpoint comparison, and the Sysala/Čermák–Sysala–Valdman plasticity references contextualize this evidence rather than define universal baselines. Natural extensions are path-dependent mechanics validation against incremental-history references, broader agreement-focused external comparisons under matched problem definitions, and reduced preconditioner-setup and multigrid costs in sparse mechanics configurations. Those extensions would test whether the coupled solver-design discipline of the present study carries over to broader nonlinear mechanics and design problems.

A Solver-Policy Diagnostics

This appendix collects supporting solver-policy diagnostics that are methodologically useful but narrower than the main benchmark narratives.

A.1 Reference-Formula and Frozen-PMG Diagnostics for 3D Mohr–Coulomb

The main 3D plasticity numerical evidence in Section 7.7 is the rank-local constitutive-AD path. The tables below add two solver-policy comparisons on the same $P_4(L_2)$ problem from (33) at $\lambda_{\mathrm{sr}} = 1.0$: a reference-formula branch-tangent assembly comparator and a frozen-preconditioner PMG variant. Both use Armijo Newton, a gradient-only stopping target $\|\nabla \mathcal{F}\|_2 < 0.01$, at most 50 Newton iterations, and PETSc relative KSP tolerance 1×10^{-1}. Here reference-formula assembly means evaluating the active constitutive branch and tangent formulas without AD, while frozen-preconditioner PMG means a variant in which the multigrid operator is not rebuilt along the nonlinear path.

Table 29: Matched constitutive-assembly versus reference-formula assembly comparison for the same PMG solver configuration on $P_4(L_2)$ at $\lambda_{\mathrm{sr}} = 1.0$. The table reports wall time, solve time, and their ratio for matched rank counts on (33).

Ranks	AD wall [s]	Ref. wall [s]	AD solve [s]	Ref. solve [s]	Wall ratio	$\|g\|_{\max}$	Status
4	1717	951	1634	890	1.805	5.84×10^{-3}	completed
8	1164	697	1107	656	1.671	5.84×10^{-3}	completed
16	608	374	575	350	1.628	5.84×10^{-3}	completed
32	300	277	284	260	1.085	5.84×10^{-3}	completed

These diagnostics support two narrower statements. First, reference-formula branch-tangent assembly has lower reported wall time than AD-branch tangent assembly in the listed matched-rank comparisons, although the gap narrows at higher rank counts. All entries in the matched reference-formula comparison are marked completed under the stated gradient stopping target in the underlying comparison summaries. Second, the frozen-preconditioner PMG profile is not the default used in the present study: it may appear favorable on completed small MPI-rank-count configurations, but three of the eight frozen-preconditioner configurations reach the maximum number of Newton iterations and therefore give less consistent convergence evidence. The rank-local constitutive-AD path remains the primary 3D plasticity realization because it uses the same local differentiable formulation as the derivative-route and converged scaling studies; the reference-formula configurations are retained to expose solver-policy sensitivity on matched discrete problems.

Table 30: Frozen-preconditioner PMG alternative on the same $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.0$ 3D plasticity problem. The status column distinguishes completed configurations from rank counts that reach the maximum number of Newton iterations before satisfying the target gradient tolerance.

Outcome and wall time

Ranks	PMG operator and tangent	Status	Wall time [s]
4	frozen PMG operator, AD branch tangent	completed	2116
4	frozen PMG operator, reference-formula branch tangent	completed	988
8	frozen PMG operator, AD branch tangent	completed	1874
8	frozen PMG operator, reference-formula branch tangent	completed	898
16	frozen PMG operator, AD branch tangent	iteration cap	1304
16	frozen PMG operator, reference-formula branch tangent	iteration cap	601
32	frozen PMG operator, AD branch tangent	completed	348
32	frozen PMG operator, reference-formula branch tangent	iteration cap	398

Newton–Krylov work

Ranks	PMG operator and tangent	Newton iters	Krylov iters	Final gradient norm
4	frozen PMG operator, AD branch tangent	24	204	6.53×10^{-3}
4	frozen PMG operator, reference-formula branch tangent	21	210	2.49×10^{-3}
8	frozen PMG operator, AD branch tangent	33	238	2.53×10^{-3}
8	frozen PMG operator, reference-formula branch tangent	27	246	9.53×10^{-3}
16	frozen PMG operator, AD branch tangent	50	196	2.21×10^{2}
16	frozen PMG operator, reference-formula branch tangent	50	177	7.35×10^{2}
32	frozen PMG operator, AD branch tangent	29	250	2.24×10^{-3}
32	frozen PMG operator, reference-formula branch tangent	50	193	8.25×10^{1}

A.2 2D Mohr–Coulomb Reference Continuation Evidence

For 2D plasticity, the reference-continuation comparison is not the primary method of the paper, but it is still useful as fixed-policy rank and timing evidence for a fixed same-mesh PMG policy under the same outer continuation logic.

Table 31: Reference continuation with the fixed PMG policy at 8 and 32 ranks. This is supporting solver evidence for a reference continuation setting, not a primary solver comparison.

Policy	Ranks	Wall time [s]	Init Krylov iters	Continuation Krylov iters	Final λ_{sr}
fixed PMG policy	8	489	58	733	1.568403
fixed PMG policy	32	198	60	829	1.568334

The two entries show fixed-policy timing evidence rather than a validation baseline: increasing from 8 to 32 ranks reduces the reported wall time from 489 s to 198 s, while the final strength-reduction values remain close at 1.568403 and 1.568334.

Code and Data Availability

A version-controlled repository containing the source code, input data, processed benchmark summaries, and figure and table generation scripts supporting this study is available at the GitHub repository `https://github.com/Beremi/jax_petsc_nonlinear_energies`. The final archival release and DOI will be cited in the submitted version.

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